SGTP: Sampling-based G for Real-Time Multi-Vehi

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***Abstract*—Autonomous multi-vehicle racing requires real-time planning of diverse competitive behaviors in intense interactions. Existing planners often struggle to balance strategic diversity and computational efficiency. To address this challenge, we propose Sampling-based Game-Theoretic Planning (SGTP), a real-time framework that combines game-theoretic reasoning with GPU-accelerated sampling of control sequences and dynamics rollouts. Sampled trajectories are ranked using a game-aware cost to capture competitive interactions and generate diverse racing behaviors. Our planner then performs feasibility selection by explicitly enforcing track-boundary and dynamic collision-avoidance constraints, ensuring safe and reliable transitions between racing strategies. Extensive simulations on challenging tracks show that SGTP achieves a 95.24% win rate and a 99.35% task-completion ratio in highly interactive races, with a mean computational time of 0.095 s over multiple iterative solving steps. We also demonstrate the successful application of SGTP in large-scale scenarios with up to 10 agents. We release our code and provide an open-source benchmark of multi-agent autonomous racing algorithms to facilitate future research. Project page: https://sgtp-racing.github.io/.**

***Index Terms*—Multi-vehicle planning, Game theory, Iterative best response, Model Predictive Control, Autonomous Racing**

I. Introduction

In Formula 1 racing, drivers continuously manage gaps to create overtaking opportunities and defend their positions. Reproducing such sustained wheel-to-wheel interaction remains challenging in autonomous racing, as it requires planners to generate multiple competitive behaviors and transition reliably between them as interactions evolve.

Despite extensive research on rule-based, optimization-based, sampling-based, and learning-based trajectory planning methods, generating competitive behaviors in multi-vehicle racing remains challenging [1]. Finite-State-Machine (FSM)-based planners [2], [3] have shown effective multi-behavior planning, but they rely on expert-designed behavior modes and heavy parameter tuning. Model Predictive Contouring Control (MPCC) [4]–[6] and Model Predictive Control (MPC) [7] explicitly model racing progress and reference velocity profiles. However, close-proximity interactions make the resulting optimization problems sensitive to initialization, leading to variable computational time. Sampling-based planners [8] can provide an effective trade-off between domain exploration and computational efficiency, but existing methods are limited to 1-2 opponent vehicles. Learning-based methods, including end-to-end policies [9] and generative models [10], [11], can represent rich interaction behaviors, but their limited interpretability and cross-scenario generalization remain concerns [12]. These limitations motivate new planning methods that leverage

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SGTP：基于采样的博弈规划实时多车辆

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摘要—自主多车辆赛车需要在紧张的交互中实时规划多样化的竞争行为。现有规划器往往难以平衡策略多样性和计算效率。为应对这一挑战，我们提出了基于采样的博弈规划（SGTP），这是一个结合博弈推理与GPU加速控制序列和动力学展开采样的实时框架。采样轨迹通过博弈感知成本进行排序，以捕获竞争交互并生成多样化的赛车行为。我们的规划器通过显式执行赛道边界和动态避碰约束进行可行性选择，确保在赛车策略之间实现安全可靠的转换。在具有挑战性的赛道上的大量仿真表明，SGTP在高度交互的比赛中实现了95.24%的胜率和99.35%的任务完成率，在多次迭代求解步骤中平均计算时间为0.095秒。我们还展示了SGTP在最多包含10个智能体的场景中的成功应用。我们发布了代码，并提供了一个多智能体自主赛车算法的开放源码基准，以促进未来研究。项目页面：https://sgtp-racing.github.io/。

索引词—多车辆规划，博弈论，迭代最佳响应，模型预测控制，自主赛车

I. 引言

在一级方程式赛车中，车手们持续管理车距以创造超车机会并捍卫自己的位置。在自主赛车中，重现这种持续的車对车交互仍然具有挑战性，因为它需要规划者生成多种竞争行为，并在交互发展过程中可靠地在它们之间进行转换。

尽管在基于规则、基于优化、基于采样和基于学习的轨迹规划方法方面进行了大量研究，但在多车辆赛车中生成具有竞争力的行为仍然具有挑战性 [1]。基于有限状态机（FSM）的规划器 [2], [3] 已显示出有效的多行为规划能力，但它们依赖于专家设计的行為模式，并且需要大量的参数调整。模型预测轮廓控制（MPCC） [4]–[6] 和模型预测控制（MPC） [7]明确地建模了赛车进程和参考速度曲线。然而，近距离交互使得生成的优化问题对初始化敏感，导致计算时间变化。基于采样的规划器 [8]可以在领域探索和计算效率之间提供有效的权衡，但现有方法仅限于1-2辆对手车辆。基于学习的方法，包括端到端策略 [9] 和生成模型 [10], [11], 可以表示丰富的交互行为，但它们的可解释性和跨场景泛化能力有限，仍然是问题 [12]。这些局限性促使人们开发新的规划方法，这些方法可以利用

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Game-Theoretic Planning
hicle Autonomous Racing

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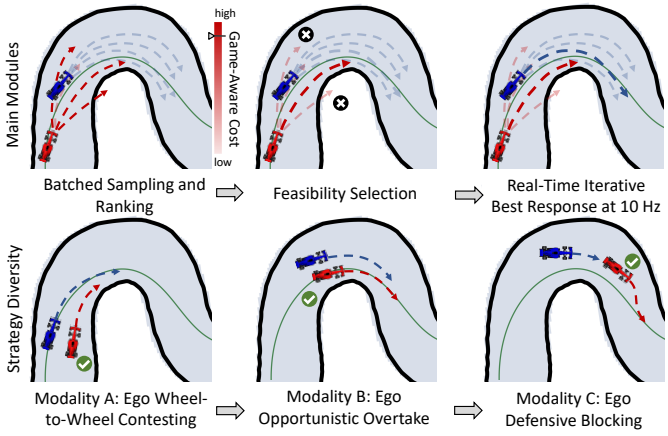


Fig. 1: **Proposed SGTP framework for multi-behavior autonomous racing.** SGTP combines GPU-accelerated sampling-based planning with game-theoretic best-response reasoning, using a new game-aware cost to favor competitive interactions and selecting the lowest-cost feasible trajectory.

interaction-aware reasoning to explore diverse strategies while ensuring reliable transitions between them.

Game theory provides a principled framework for interaction-aware reasoning and multi-strategy planning [13]–[15]. Accordingly, multi-vehicle racing can be formulated as a Generalized Nash Equilibrium Problem [16]–[18] and is commonly solved using Iterative Best Response (IBR), which optimizes each vehicle's strategy while holding those of the other vehicles fixed [17], [19], [20]. However, consistently obtaining feasible and high-quality solutions to each best-response subproblem within the available online computation time remains a major bottleneck [17], [18]. Although MPC is widely adopted for its ability to explicitly handle system and safety constraints [17], [20], the resulting optimization problems are highly nonconvex [20]; consequently, both solution quality and computational time are highly sensitive to initialization [21], particularly in close-range racing scenarios [20]. As a result, MPC-based best-response solvers exhibit initialization-dependent or unpredictable convergence, and impose substantial online computational burdens [8], [17]. Model Predictive Path Integral (MPPI)-based planners [22]–[24] have emerged as GPU-accelerated sampling-based alternatives for real-time trajectory planning. However, they typically handle collision avoidance through soft cost penalties rather than explicit feasibility checks, which can lead to infeasible maneuvers in close-range wheel-to-wheel racing. Despite the potential of integrating MPPI with IBR and tailored game-theoretic cost functions to alleviate existing limitations, this combination remains unexplored.

We propose a **S**ampling-based **G**ame-**T**heoretic **P**lanning

1

博弈论规划
多车辆自主赛车

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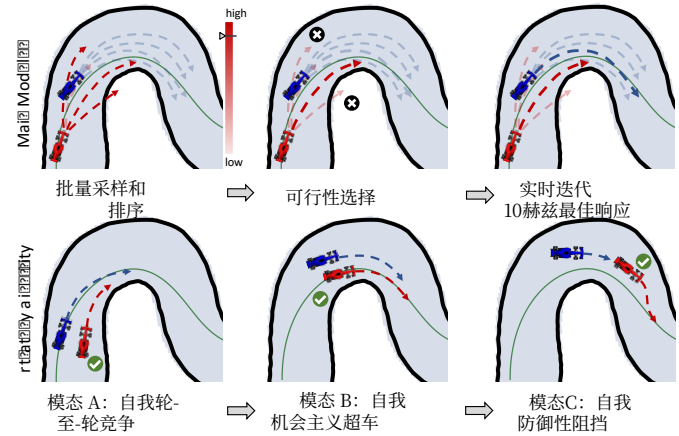


图1：**为多行为自主赛车提出的SGTP框架。** SGTP结合GPU加速采样规划与博弈论最佳响应推理，使用新的博弈感知成本来优先考虑竞争交互，并选择最低成本的可行轨迹。

交互感知推理以探索多样化策略，同时确保它们之间可靠转换。

博弈论为交互感知推理和多策略规划提供了一种原则性框架 [13]–[15]。因此，多车辆赛车可以表述为一个广义纳什均衡问题 [16]–[18]，通常使用迭代最佳响应（IBR）来解决，该算法在固定其他车辆策略的同时优化每个车辆的策略 [17], [19], [20]。然而，在可用的在线计算时间内始终如一地获得每个最佳响应子问题的可行且高质量的解决方案仍然是一个主要瓶颈 [17], [18]。尽管模型预测控制（MPC）因其能够显式处理系统和安全约束而得到广泛应用 [17], [20], ，但产生的优化问题高度非凸 [20]；因此，解决方案质量和计算时间对初始化高度敏感 [21]，特别是在近距离赛车场景中 [20]。结果，基于MPC的最佳响应求解器表现出初始化相关或不确定的收敛性，并带来巨大的在线计算负担 [8], [17]。模型预测路径积分（MPPI）规划器 [22]–[24] 已成为GPU加速采样为实时轨迹规划提供的替代方案。然而，它们通常通过软成本惩罚而不是显式可行性检查来处理碰撞避免，这可能导致近距离车对车赛车中的不可行操作。尽管将MPPI与IBR和定制的博弈论成本函数相结合以缓解现有限制具有潜力，但这种组合仍未被探索。

我们提出了一种**S**ampling-based **G**ame-**T**heoretic **P**lanning

(**SGTP**) framework for competitive multi-vehicle racing that generates diverse racing strategies in real time. SGTP integrates IBR by reformulating each best-response sub-problem as GPU-parallel control sampling, followed by dynamics roll-out and trajectory ranking. A game-aware cost encourages competitive interactions, while feasibility selection explicitly enforces track-boundary and collision-avoidance constraints, enabling reliable transitions between strategies (Fig. 1). Existing evaluation frameworks often rely on simplified maps and lack interaction-rich benchmarks, limiting the assessment of planners in highly competitive racing scenarios [18], [20]. To address this gap, we release our SGTP planner, its benchmarks and the evaluation platform as open source. The main contributions of this letter are summarized as follows:

1) **SGTP: A Real-Time, Sampling-Based, IBR Trajectory Planning Framework.** SGTP performs GPU-parallel batched rollout generation and cost-based selection to approximate unilateral best responses, followed by an additional ego response after opponent prediction. Our design enables multiple competitive behaviors in intense interactions, while achieving real-time planning with a mean computation time of **0.095s**.
2) **Game-Aware Trajectory Cost with Feasibility Selection.** A new game-aware cost evaluates rollout trajectories based on relative progress, contest status, blocking alignment, and safety margins, thereby promoting diverse competitive behaviors. Combined with feasibility selection over the ranked rollouts, SGTP achieves a **95.24%** win rate and a **99.35%** completion ratio in long-duration competitive racing.
3) **Open-Source Benchmark of Multi-Agent Autonomous Racing.** We release an open benchmark with SGTP, multiple literature baselines, and several track layouts, to evaluate trajectory planning methods in multi-vehicle autonomous racing.

II. SGTP Algorithm

a) Preliminaries and Definitions: Let $\mathcal{I} = \{1, \dots, n_v\}$ denote the set of racing vehicles, where n_v is the number of vehicles. For each vehicle $i \in \mathcal{I}$, $\mathcal{I}_{-i} = \mathcal{I} \setminus \{i\}$ denotes the set of opponent vehicles. Let t denote the current discrete planning step. The joint multi-vehicle state at time t is defined as $X_t = \{\mathbf{x}_{i,t}\}_{i\in\mathcal{I}}$, where $\mathbf{x}_{i,t} = [x_{i,t}, y_{i,t}, \psi_{i,t}, v_{i,t}]^\top$ and $\mathbf{u}_{i,t} = [a_{i,t}, \delta_{i,t}]^\top$ denote the state and control input, respectively. Here, $x_{i,t}$ and $y_{i,t}$ are the vehicle position, $\psi_{i,t}$ is the heading angle, $v_{i,t}$ is the speed, $a_{i,t}$ is the longitudinal acceleration, and $\delta_{i,t}$ is the steering angle. The vehicle dynamics are written as $\mathbf{x}_{i,t+1} = \boldsymbol{f}(\mathbf{x}_{i,t}, \mathbf{u}_{i,t})$, where $\boldsymbol{f}(\cdot)$ follows the kinematic bicycle model in [25]. The sampled control sequence $\boldsymbol{U}_{i,t}$ is $[\mathbf{u}_{i,t}, \dots, \mathbf{u}_{i,t+H-1}]$, and the corresponding trajectory $\boldsymbol{\xi}_{i,t}$ is denoted by $[\mathbf{x}_{i,t}, \dots, \mathbf{x}_{i,t+H}]$, where H is the prediction horizon. For the opponent set $\mathcal{I}_{-i}$, the predicted trajectories are $\widehat{\boldsymbol{\Xi}}_{-i,t} = \{\hat{\boldsymbol{\xi}}_{j,t}\}_{j\in\mathcal{I}_{-i}}$, where $\hat{\boldsymbol{\xi}}_{j,t}$ denotes the predicted trajectory of vehicle j. For a given racetrack, we compute a minimum-curvature raceline $\mathcal{G} = \left\{(\mathbf{x}_j^r, w_j^L, w_j^R)\right\}_{j=1}^{N_{\text{trk}}}$, where N_{trk} is the number of waypoints, $\mathbf{x}_j^r$ is the raceline state trajectory, w_j^L and w_j^R are the left and right track widths. The raceline generation follows [26]. By interpolating these

(**SGTP**)框架，用于竞争性多车辆赛车，可在实时中生成多样化的赛车策略。SGTP通过将每个最佳响应子问题重新表述为GPU并行控制Sampling，然后进行动力学Roll-out和轨迹排序，整合了IBR。博弈感知成本鼓励竞争交互，而可行性选择明确强制赛道边界和避撞约束，从而在策略之间实现可靠转换（图1）。现有的评估框架通常依赖于简化的地图，缺乏丰富的交互基准，限制了在高度竞争性赛车场景中对规划器的评估 [18], [20]。为解决这一差距，我们发布了我们的SGTP规划器、其基准和评估平台作为开源软件。本信的主要贡献总结如下：

1) **SGTP：一个实时、基于采样的IBR轨迹规划框架。** SGTP执行GPU并行批量滚动生成和基于成本的选取来近似单边最佳响应，随后在对手预测后进行额外的自我响应。我们的设计在紧张的交互中实现了多种竞争行为，同时以均值为**0.095s**实现了实时规划。

2) **博弈感知轨迹成本与可行性选择。** 一种新的博弈感知成本基于相对进度、竞赛状态、阻挡对齐和安全裕度来评估 rollout 轨迹，从而促进多样化的竞争行为。结合对排序后的 rollout 的可行性选择，SGTP 在长时竞技赛车中实现了 **95.24%** 的胜率和 **99.35%** 的完成比率。

3) **多智能体自主赛车开源基准。** 我们发布了一个基于SGTP的开源基准，包含多个文献基线以及多种赛道布局，用于评估多车辆自主赛车中的轨迹规划方法。

II. SGTP 算法

a) 前提与定义：令 $\mathcal{I} = \{1, \dots, n_v\}$ 表示赛车集合，其中 n_v 是车辆数量。对于每辆车辆 $i \in \mathcal{I}$，$\mathcal{I}_{-i} = \mathcal{I} \setminus \{i\}$ 表示对手车辆集合。令 t 表示当前离散规划步长。时间 t 的联合多车辆状态定义为 $X_t = \{\mathrm{x}_{i,\ t}\}_{i\in\mathcal{I}}$，其中 $\mathrm{x}_{i,\ t} = [x_{i,t}, y_{i,t}, \psi_{i,t}, v_{i,t}]^\top$ 和 $\mathrm{u}_{i,\ t} =, [a_{i,t}, \delta_{i,t}]^\top$ 分别表示状态和控制输入。这里，$x_{i,\ t}$ 和 $y_{i,\ t}$ 是车辆位置，$\psi_{i,\ t}$ 是航向角，$v_{i,\ t}$ 是速度，$a_{i,\ t}$ 是纵向加速度，而 $\delta_{i,\ t}$ 是转向角。车辆动力学表示为 $\mathrm{x}_{i,\ t+1} = \boldsymbol{f}(\mathrm{x}_{i,\ t}, \mathrm{u}_{i,\ t})$，其中 $\boldsymbol{f}(\cdot)$ 遵循 [25] 中的运动学自行车模型。采样控制序列 $\boldsymbol{U}_{i,\ t}$ 是 $[\mathrm{u}_{i,\ t}, \dots, \mathrm{u}_{i,\ t+H-1}]$，相应的轨迹 $\boldsymbol{\xi}_{i,\ t}$ 表示为 $[\mathrm{x}_{i,\ t}, \dots, \mathrm{x}_{i,\ t+H}]$，其中 H 是预测范围。对于对手集合 $\mathcal{I}_{-i}$，预测轨迹 $\widehat{\ }\ \widehat{\ }\ \widehat{\ }$ 是 $\boldsymbol{\Xi}_{-i,\ t} = \{\boldsymbol{\xi}_{j,t}\}_{j\in\mathcal{I}_{-i}}$，其中 $\boldsymbol{\xi}_{j,\ t}$ 表示车辆 j 的预测轨迹。对于给定赛道，我们计算一条最小曲率赛车线 $\mathcal{G} = \left\{(\mathrm{x}_j^r,\ w_j^L,\ w_j^R)\right\}_{j=1}^{N_{\text{trk}}}$，其中 N_{trk} 是航点数量，x_j^r 是赛车线状态轨迹，w_j^L 和 w_j^R 是左右赛道宽度。赛车线生成遵循 [26]。通过插值这些

Algorithm 1 Sampling-Based Game-Theoretic Planning

1: **Inputs:** Current states $\{\mathbf{x}_{i,t}\}_{i\in\mathcal{I}}$, raceline $\mathcal{G}$, horizon H, samples K, IBR iterations L_{IBR}, ego vehicle index e
2: **Outputs:** Planned trajectories for all agents $\widehat{\boldsymbol{\Xi}} = \{\boldsymbol{\xi}_i^\star\}_{i\in\mathcal{I}}$
3: Initialize $\{\boldsymbol{\xi}_i^0\}_{i\in\mathcal{I}}$ without game-aware costs. ▷ Distance-based obstacle avoidance
4: **for** $\ell = 1, \ldots, L_{\text{IBR}}$ **do**
5: **for** each vehicle $i \in \mathcal{I}$ **do**
6: Set $\widehat{\boldsymbol{\Xi}}_{-i}^{\ell} \leftarrow \{\hat{\boldsymbol{\xi}}_j^{\ell-1}\}_{j\in\mathcal{I}_{-i}}$ ▷ Opponent predictions
7: Generate $\mathcal{C}_i^\ell = \{(\boldsymbol{\xi}_{i,t}^{(k,\ell)}, \boldsymbol{U}_{i,t}^{(k,\ell)})\}_{k=1}^K$
8: Compute cost $c_i^{(k,\ell)}$ via (9) $\forall$ candidate k in $\mathcal{C}_i^\ell$
9: Compute feasibility indicators $h_i^{(k,\ell)}$ via (12)
10: Apply selection $k_i^{\star,\ell} \leftarrow \arg\min_{k:h_i^{(k,\ell)}=1} c_i^{(k,\ell)}$
11: Update $\boldsymbol{\xi}_i^\ell \leftarrow \boldsymbol{\xi}_i^{(k^{\star,\ell})}$ and $\overline{\boldsymbol{U}}_i^{\ell+1} \leftarrow \boldsymbol{U}_i^{(k^{\star,\ell})}$
12: **end for**
13: **end for**
14: $k_e^* = \arg\min_{k:h_e^{(k)}=1} c_e^{(k)}$ ▷ Extra ego response

arc-length-parameterized waypoints, $\mathcal{G}$ defines continuous reference maps $\mathbf{x}^r(s)$, $w^L(s)$, and $w^R(s)$, which are used for reference tracking and track-boundary evaluation. The Frenet progress $s_{i,t}$ along the raceline and lateral offset $d_{i,t}$ of vehicle i are then computed as: $s_{i,t} = s(\mathbf{x}_{i,t}; \mathcal{G})$, $d_{i,t} = d(\mathbf{x}_{i,t}; \mathcal{G})$.

b) Algorithm Formulation: Algorithm 1 summarizes our SGTP framework. The trajectories of all vehicles are initialized using simple distance-based collision avoidance (line 3). The algorithm loops over the IBR iterations and vehicles (lines 4-5). At each IBR iteration, every vehicle $i \in \mathcal{I}$ computes an approximate best response, while the predicted trajectories of its opponents are held fixed at their preceding-iteration solutions (line 6). To compute the best response, we sample K candidate control sequences and propagate them in parallel through the vehicle dynamics (line 7). Let $\overline{\boldsymbol{U}}_{i,t}^{\ell} = [(\overline{\mathbf{u}}_{i,t}^{\ell})^\top, \ldots, (\overline{\mathbf{u}}_{i,t+H-1}^{\ell})^\top]^\top$ denote the nominal control sequence of vehicle i at IBR iteration ℓ, used as a reference for candidate sampling. $\overline{\boldsymbol{U}}_{i,t}^{\ell}$ is initialized to zero upon reset and then warm-started with the control sequence $\overline{\boldsymbol{U}}_{i,t}^{\ell-1}$ from the previous iteration. For each of the K candidate trajectories and each prediction step τ, an independent Gaussian perturbation is sampled as $\boldsymbol{\epsilon}_{i,t+\tau}^{(k,\ell)} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_u)$, $\boldsymbol{\Sigma}_u = \text{diag}(\sigma_a^2, \sigma_\delta^2)$, where σ_a and σ_δ are the standard deviations of the acceleration and steering perturbations. The perturbed control input is generated and projected onto an admissible control set $\mathcal{U}$:

$$\mathbf{u}_{i,t+\tau}^{(k,\ell)} = \Pi_{\mathcal{U}}\left(\widetilde{\mathbf{u}}_{i,t+\tau}^{(k,\ell)}\right), \text{ where } \widetilde{\mathbf{u}}_{i,t+\tau}^{(k,\ell)} = \overline{\mathbf{u}}_{i,t+\tau}^{\ell} + \boldsymbol{\epsilon}_{i,t+\tau}^{(k,\ell)}.$$

Here $\Pi_{\mathcal{U}}(\cdot)$ applies clipping to acceleration and steering limits. Starting from $\mathbf{x}_{i,t}^{(k,\ell)} = \mathbf{x}_{i,t}$, we propagate all sampled control sequences in parallel through the dynamics $\boldsymbol{f}(\cdot)$, generating the finite rollout set $\mathcal{C}_{i,t}^{\ell} = \{(\boldsymbol{\xi}_{i,t}^{(k,\ell)}, \boldsymbol{U}_{i,t}^{(k,\ell)})\}_{k=1}^K$ (line 7).

Given the fixed opponent predictions $\widehat{\boldsymbol{\Xi}}_{-i}^{\ell}$, the sampled rollouts are ranked by our game-aware cost (described in Sec. III-A) and checked for feasibility to enforce track-boundary and collision-avoidance constraints (lines 8-9). The lowest-cost feasible rollout is selected in line 10 as:

$$k_i^{\star,\ell} = \arg\min_{k:h_i^{(k,\ell)}=1} c_i\left(\boldsymbol{\xi}_{i,t}^{(k,\ell)}, \widehat{\boldsymbol{\Xi}}_{-i,t}^{\ell}, \mathcal{G}\right), \tag{1}$$

算法1 基于采样的博弈规划

1: **输入：** 当前状态 $\{\mathbf{x}_{i,t}\}_{i\in\mathcal{I}}$，赛道线 $\mathcal{G}$，视野 H，样本 K，IBR迭代次数 L_{IBR}，自我车辆索引 e
2: **输出：** 所有智能体的规划轨迹 $\widehat{\boldsymbol{\Xi}} = \{\boldsymbol{\xi}_i^\star\}_{i\in\mathcal{I}}$
3: 初始化 $\{\boldsymbol{\xi}_i^0\}_{i\in\mathcal{I}}$ 而不考虑博弈感知成本。 ▷ 距离-基于障碍物避让
4: **为** $\ell = 1, \ldots, L_{\text{IBR}}$ **执行**
5: **为** 每辆车 $i \in \mathcal{I}$ **执行**
6: 设置 $\widehat{\boldsymbol{\Xi}}_{-i}^{\ell} \leftarrow \{\hat{\boldsymbol{\xi}}_j^{\ell-1}\}_{j\in\mathcal{I}_{-i}}$ ▷ 对手预测
7: 生成 $\mathcal{C}_i^\ell = \{(\boldsymbol{\xi}_{i,t}^{(k,\ell)}, \boldsymbol{U}_{i,t}^{(k,\ell)})\}_{k=1}^K$
8: 计算成本 $c_i^{(k,\ell)}$ 通过 (9) $\forall$候选 k 在 $\mathcal{C}_i^\ell$
9: 计算可行性指标 $h_i^{(k,\ell)}$ 通过 (12)
10: 应用选择 $k_i^{\star,\ell} \leftarrow \arg\min_{k:h_i^{(k,\ell)}=1} c_i^{(k,\ell)}$
11: 更新 $\boldsymbol{\xi}_i^\ell \leftarrow \boldsymbol{\xi}_i^{(k^{\star,\ell})}$ 和 $\overline{\boldsymbol{U}}_i^{\ell+1} \leftarrow \boldsymbol{U}_i^{(k^{\star,\ell})}$
12: **end for**
13: **结束for**
14: $k_e^* = \arg\min_{k:h_e^{(k)}=1} c_e^{(k)}$ ▷ 额外的自我响应

弧长参数化的航点，$\mathcal{G}$ 定义了连续参考地图 $\mathbf{x}^r(s)$，$w^L(s)$ 和 $w^R(s)$，这些地图用于参考跟踪和赛道边界评估。赛车线上的弗内特进度 $s_{i,t}$ 和车辆 i 的横向偏移 $d_{i,t}$ 计算为：$s_{i,t} = s(\mathbf{x}_{i,t}; \mathcal{G})$ $d_{i,t} = d(\mathbf{x}_{i,t}; \mathcal{G})$。

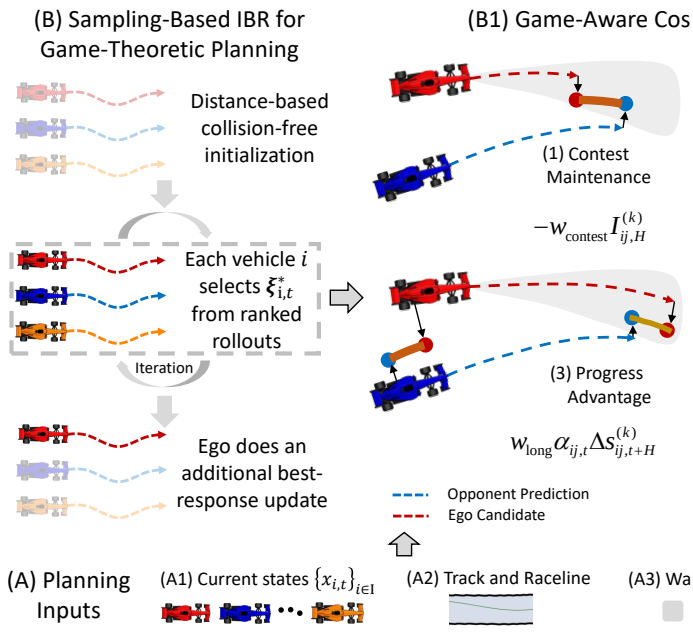
b) 算法公式化：算法1总结了我们的SGTP框架。所有车辆的轨迹使用简单的基于距离的避撞初始化（第3行）。该算法遍历IBR迭代和车辆（第4-5行）。在每个IBR迭代中，每个车辆 $i \in \mathcal{I}$计算一个近似最佳响应，同时其对手的预测轨迹保持在其前一个迭代解的固定值（第6行）。为了计算最佳响应，我们采样 K 候选控制序列，并将它们并行通过车辆动力学（第7行）。令 $\overline{\boldsymbol{U}}_{i,t}^{\ell} = [(\mathrm{u}_{i,t}^{\ell})^\top, \ldots, (\mathrm{u}_{i,t+H-1}^{\ell})^\top]^\top$ 表示车辆 i 在IBR迭代 ℓ中的标称控制序列，用作候选采样的参考。$\overline{\boldsymbol{U}}_{i,t}^{\ell}$ 在重置时初始化为零，然后使用来自前一个迭代的控制序列 $\overline{\boldsymbol{U}}_{i,t}^{\ell-1}$ 进行热启动。对于每个候选轨迹和每个预测步 τ，采样一个独立的正态扰动作为 $\boldsymbol{\epsilon}_{i,t+\tau}^{(k,\ell)} \sim \mathcal{N}(0, \boldsymbol{\Sigma}_u)$ $\boldsymbol{\Sigma}_u = \text{diag}(\sigma_a^2, \sigma_\delta^2)$，其中 σ_a 和 σ_δ 是加速度和转向扰动的标准差。扰动控制输入被生成并投影到可接受的控制集 $\mathcal{U}$：

$$\mathbf{u}_{i,t+\tau}^{(k,\ell)} = \Pi_{\mathcal{U}}\left(\widetilde{\mathbf{u}}_{i,t+\tau}^{(k,\ell)}\right), \text{ where } \widetilde{\mathbf{u}}_{i,t+\tau}^{(k,\ell)} = \overline{\mathbf{u}}_{i,t+\tau}^{\ell} + \boldsymbol{\epsilon}_{i,t+\tau}^{(k,\ell)}.$$

这里 $\Pi_{\mathcal{U}}(\cdot)$ 将加速度和转向限制进行裁剪。从 $\mathrm{x}_i^{(k,\ell)}{}_{,t} = \mathrm{x}_{i,t}$ 开始，我们并行地将所有采样的控制序列通过动力学 $\boldsymbol{f}(\cdot)$ 传播，生成有限策略集 $\mathcal{C}_{i,t}^{\ell} = \{(\boldsymbol{\xi}_{i\,t}^{(k,\ell)}, \boldsymbol{U}_{(k,\ell)\,i,t})\}_{k=1}^K$（第7行）。

给定固定的对手预测 $\widehat{\boldsymbol{\Xi}}_{-i}^{\ell}$，所采样的轨迹根据我们的博弈感知成本（详见第III-A 节）进行排序，并检查可行性以强制执行赛道边界和避撞约束（第 8-9 行）。最低成本的可行轨迹在第 10 行被选中，其表示为：III-A）并检查可行性以强制执行赛道边界和避撞约束（第 8-9 行）。最低成本的可行轨迹在第 10 行被选中，其表示为：8-9）。最低成本的可行轨迹在第 10 行被选中，其表示为：10 作为：

$$k_i^{\star,\ell} = \arg\min_{k:h_i^{(k,\ell)}=1} c_i\left(\boldsymbol{\xi}_{i,t}^{(k,\ell)}, \widehat{\boldsymbol{\Xi}}_{-i,t}^{\ell}, \mathcal{G}\right), \tag{1}$$



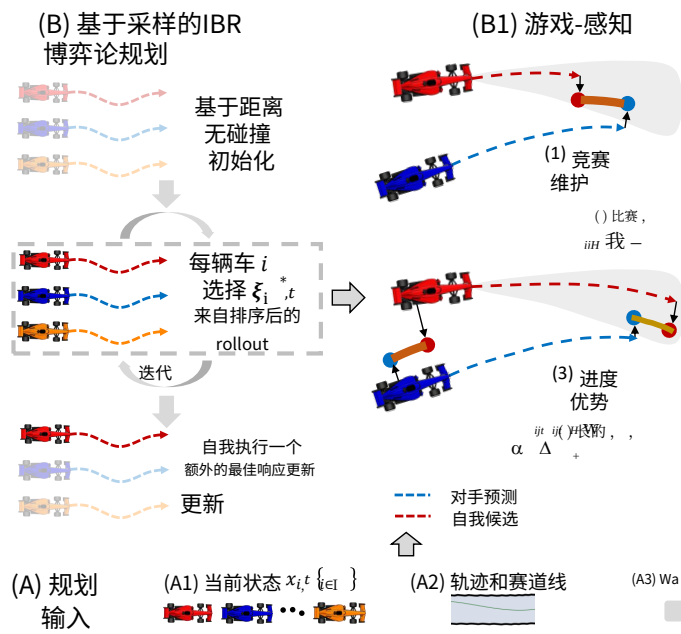


Fig. 2: **Proposed Sampling-Based Game-Theoretic Planning** and ranks them using a game-aware cost evaluated against opp response, after which feasibility selection enforces the track-bo ego vehicle then performs an additional best-response update.

Here, $k_i^{\star,\ell}$ denotes the selected candidate index, and $h_i^{(k,\ell)} = 1$ indicates that candidate k satisfies both track-boundary and dynamic vehicle collision-avoidance constraints.

The selected trajectory and control sequence warm start the next IBR iteration (line 11). After the prescribed number of iterations L_{IBR}, the ego vehicle performs one additional best-response update with the final opponents' predictions (line 14).

III. Sampling-Based Game-Theoretic Planning

As shown in Fig. 2, SGTP performs best-response planning by ranking rollouts (generated in parallel on the GPU) using a game-aware cost, and selecting the lowest-cost feasible rollout.

A. Game-Aware Cost for Diverse Behaviors

We compute our game-aware cost by evaluating a pairwise interaction term between each candidate trajectory and the predicted opponent trajectories. For the candidate trajectory $\boldsymbol{\xi}_{i,t}^{(k,\ell)}$ of vehicle i and the predicted trajectory of opponent j, the pairwise game-aware cost is defined as follows:

$$c^{(k)}_{\text{game},ij} = c^{(k)}_{\text{contest},ij} + c^{(k)}_{\text{long},ij} + c^{(k)}_{\text{block},ij} + c^{(k)}_{\text{safety},ij}. \quad (2)$$

The four cost terms represent contest-state maintenance c_{contest}, longitudinal advantage c_{long}, blocking alignment c_{block}, and safety separation c_{safety}, respectively.

The key design insight is to progressively shape the interaction through complementary cost terms: the contest term preserves rollouts that remain within a meaningful racing window, the longitudinal term promotes forward advantage, the blocking term encourages defensive positioning, and the safety term penalizes insufficient separation. The total game-aware cost for candidate k is obtained by summing the pairwise costs over all opponents: $c^{(k)}_{\text{game},i} = \sum_{j\in\mathcal{I}_{-i}} c^{(k)}_{\text{game},ij}$, allowing each opponent prediction to independently contribute to the ranking of candidate trajectories.

图2：**提出的基于采样的博弈规划**，使用博弈感知成本对输出进行排序，该成本针对对手响应进行评估，之后可行性选择强制执行赛道-自我车辆，然后执行额外的最佳响应更新。

在此处，$k^{\star,\ell}_i$ 表示所选候选索引，而 $h_i^{(k,\ell)} = 1$表示候选 k 满足赛道边界和动态车辆避撞约束。

选定的轨迹和控制序列为下一个IBR迭代提供预热（行11）。在规定的迭代次数 L_{IBR}后，自我车辆执行一次额外的最佳响应更新，使用最终的对手预测（行14）。

III. 基于采样的博弈规划

如图2所示，SGTP通过使用博弈感知成本对并行在GPU上生成的输出进行排序，执行最佳响应规划，并选择最低成本的可行输出。

A. 博弈感知成本用于多样化行为

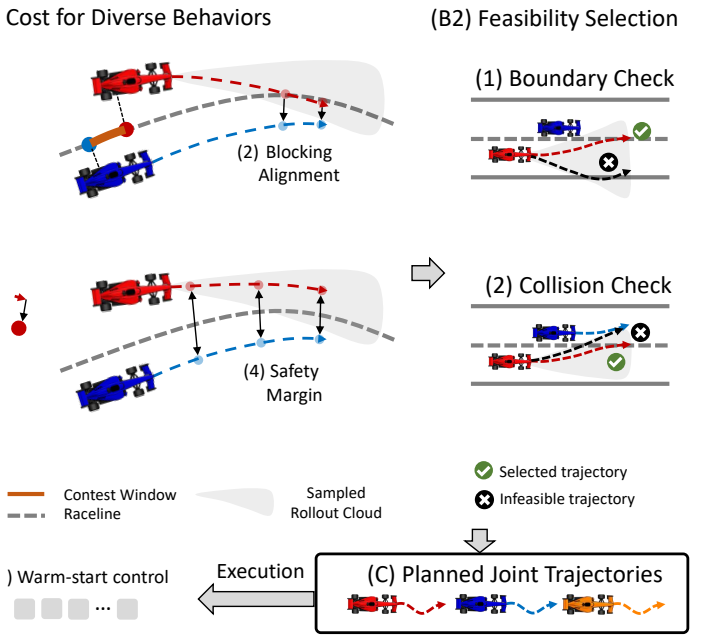
我们通过评估每个候选轨迹与预测的对手轨迹之间的成对交互项来计算我们的博弈感知成本。对于车辆 i 的候选轨迹 $\boldsymbol{\xi}_{i,t}^{(k,\ell)}$，以及对手 j 的预测轨迹，成对博弈感知成本定义如下：

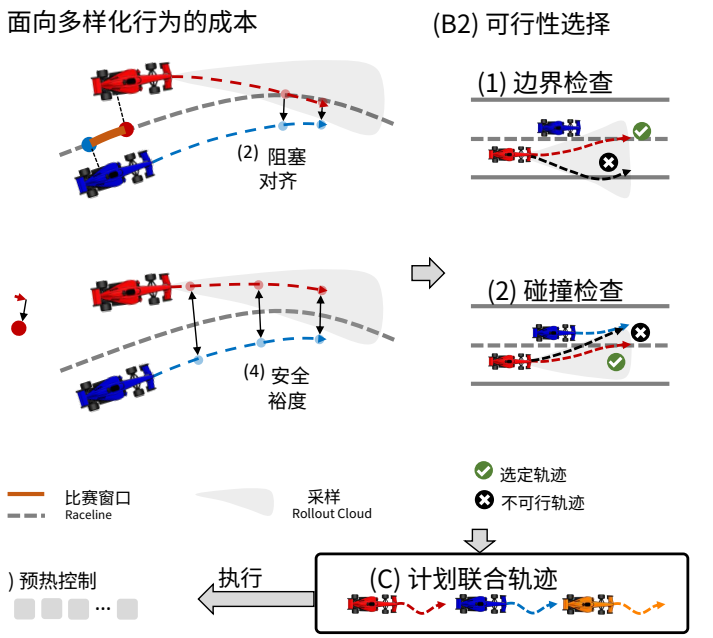
$$c^{(k)}_{\text{game},ij} = c^{(k)}_{\text{contest},ij} + c^{(k)}_{\text{long},ij} + c^{(k)}_{\text{block},ij} + c^{(k)}_{\text{safety},ij}. \quad (2)$$

我们的成本项分别表示竞争状态维护c_{contest}、纵向优势 c_{long}、阻挡对齐c_{block}和安全性分离 c_{safety}。

关键设计思路是通过互补成本项逐步塑造交互：竞赛项保留仍处于有意义比赛窗口内的 rollout，纵向项促进前进优势，阻挡项鼓励防御站位，安全项惩罚分离不足。候选 k 的总游戏感知成本通过对所有对手的成对成本求和获得：$c^{(k)}_{\text{游戏},i} = \sum_{j\in\mathcal{I}_{-i}} c^{(k)}_{\text{游戏},ij}$，允许每个对手预测独立地为候选轨迹的排序做出贡献。

3





ng (SGTP) framework. SGTP first generates sampled rollouts pponent predictions. During IBR, each vehicle computes a best -boundary and inter-vehicle collision-avoidance constraints. The .

ng (SGTP) 框架。 SGTP 首先生成采样轨迹 pponent 预测。在 IBR 过程中，每辆车计算最佳边界和车辆间碰撞避免约束。

1) Contest-State Maintenance Cost: The contest-state term rewards candidate trajectories whose terminal progress gaps relative to the predicted opponent trajectories remain within the longitudinal interaction range. Let $s_{i,t+H}^{(k)}$ denote the terminal Frenet progress of ego candidate k, and let $\hat{s}_{j,t+H}$ denote the corresponding predicted progress of opponent j. The pairwise terminal progress gaps are defined as: $\Delta s_{ij,t+H}^{(k)} = s_{i,t+H}^{(k)} - \hat{s}_{j,t+H}$. Based on these gaps, the contest indicator for candidate k and opponent j is:

$$I_{ij,H}^{(k)} = \mathbb{I}\left[|\Delta s_{ij,t+H}^{(k)}| < s_{\text{contest}}\right],$$

where s_{contest} denotes the longitudinal contest range. The corresponding pairwise contest-state cost is:

$$c_{\text{contest},ij}^{(k)} = -w_{\text{contest}} I_{ij,H}^{(k)}. \tag{3}$$

where w_{contest} is a tunable weight. The core insight is to encourage sampled rollouts to interact closely with opponents' predicted trajectories within a meaningful competitive window.

2) Longitudinal Advantage Cost: The longitudinal advantage term encourages candidate trajectories that achieve a terminal progress advantage over opponent j. It is defined as:

$$c_{\text{long},ij}^{(k)} = -w_{\text{long}} \alpha_{ij,t} \Delta s_{ij,t+H}^{(k)}, \tag{4}$$

where w_{long} is a scalar weight, and $\Delta s_{ij,t+H}^{(k)}$ denotes the terminal progress gap. A larger positive terminal gap reduces the cost, thereby favoring candidates that move ahead of the opponent. The adaptive weight $\alpha_{ij,t}$ adjusts this progress reward according to the current interaction relevance:

$$\alpha_{ij,t} = \left(1 + |\Delta s_{ij,t}|(s_{\text{contest}} + \epsilon)^{-1}\right)^{-1}.$$

Here ϵ is a small positive constant. A larger $\alpha_{ij,t}$ assigns greater importance to terminal progress gains when the two vehicles are close. As their progress gap increases, $\alpha_{ij,t}$ decreases the reward due to weaker interaction relevance.

1) 竞赛状态维护成本：竞赛状态项奖励那些相对于预测的对手轨迹，终端进度差距仍处于纵向交互范围内的候选轨迹。令 $s_i^{(k)}$，$t+H$ 表示自我候选 k 的终端弗内特进度，令 $\hat{s}_{j,t+H}$ 表示对手 j 的对应预测进度。成对终端进度差距定义为：$\Delta s_{ij}^{(k)}$，$t+H$ = $s_{i\,t+H}^{(k)} - s_{j\,t+H}$。

基于这些差距，候选 k 和对手 j 的竞赛指标为：

$$I_{ij,H}^{(k)} = \mathbb{I}\left[|\Delta s_{ij,t+H}^{(k)}| < s_{\text{contest}}\right],$$

其中 s_{contest} 表示纵向竞赛范围。相应的成对竞赛状态成本为：

$$c_{\text{contest},ij}^{(k)} = -w_{\text{contest}} I_{ij,H}^{(k)}. \tag{3}$$

其中 w_{contest} 是一个可调权重。核心思想是鼓励采样轨迹与对手预测轨迹在有意义竞争窗口内紧密交互。

2) 纵向优势成本：纵向优势项鼓励候选轨迹在终点进度上优于对手 j。其定义为：

$$c_{\text{long},ij}^{(k)} = -w_{\text{long}} \alpha_{ij,t} \Delta s_{ij,t+H}^{(k)}, \tag{4}$$

其中 w_{long} 是一个标量权重，$\Delta s_{ij}^{k}{}_{,t+H}$ 表示终点进度差距。较大的正终点差距会降低成本，从而倾向于领先对手的候选轨迹。自适应权重$\alpha_{ij,t}$ 根据当前交互相关性调整此进度奖励：

$$\alpha_{ij,t} = \left(1 + |\Delta s_{ij,t}|(s_{\text{contest}} + \epsilon)^{-1}\right)^{-1}.$$

Here ϵ 是一个小的正常数。一个更大的 $\alpha_{ij,t}$ 在两辆车靠近时赋予终端进度增益更大的重要性。随着它们进度差距的增加，$\alpha_{ij,t}$由于交互相关性较弱而减少奖励。

3) Blocking Alignment Cost: The blocking alignment term encourages candidate trajectories whose lateral positions align with the predicted position of opponent j near the end of the prediction horizon. Let $\mathcal{T}_{\text{tail}}$ denote the set of horizon indices within the final μ_{tail} portion of the prediction horizon. The mean lateral offsets of candidate k and opponent j over this final portion of the horizon are defined as follows:

$$\bar{d}_i^{(k)} = \frac{1}{|\mathcal{T}_{\text{tail}}|} \sum_{\tau \in \mathcal{T}_{\text{tail}}} d_{i,t+\tau}^{(k)}, \; \bar{d}_j = \frac{1}{|\mathcal{T}_{\text{tail}}|} \sum_{\tau \in \mathcal{T}_{\text{tail}}} \hat{d}_{j,t+\tau},$$

where $\hat{d}_{j,t+\tau}$ denotes the predicted lateral offset of opponent j. The corresponding tail lateral gap is: $\Delta d_{ij,\text{tail}}^{(k)} = \left| \bar{d}_i^{(k)} - \bar{d}_j \right|$. Using this gap, the blocking alignment cost is formulated as:

$$c_{\text{block},ij}^{(k)} = -\gamma_{\text{block},ij} \left(1 + \Delta d_{ij,\text{tail}}^{(k)}\right)^{-1}. \tag{5}$$

Here, $\gamma_{\text{block},ij} = w_{\text{block}} \mathbb{I}[\Delta s_{ij,t} > s_{\text{role}}] \, \mathbb{I}[\Delta s_{ij,t} < s_{\text{contest}}]$ activates the blocking cost when vehicle i has a longitudinal advantage of at least s_{role} while remaining within the contest range s_{contest}, where $\Delta s_{ij,t} = s_{i,t} - \hat{s}_{j,t}$. The scalar w_{block} is the weight. Such lateral alignment reduces the space available to the trailing vehicle and promotes defensive blocking.

4) Safety Cost: The safety term penalizes candidate rollouts whose predicted separation from an opponent falls below the desired threshold g_{safe}. For candidate k and opponent j, the minimum time-aligned distance is:

$$g_{\min,ij}^{(k)} = \min_{\tau=1,\ldots,H} \left\| \mathbf{p}_{i,t+\tau}^{(k)} - \hat{\mathbf{p}}_{j,t+\tau} \right\|_2, \tag{6}$$

where $\mathbf{p}_{i,t+\tau}^{(k)}$ and $\hat{\mathbf{p}}_{j,t+\tau}$ denote the position associated with candidate k and the predicted position of opponent j at time $t+\tau$, respectively. The corresponding pairwise safety cost is:

$$c_{\text{safety},ij}^{(k)} = w_{\text{safety}} \left[\max\left(0, g_{\text{safe}} - g_{\min,ij}^{(k)}\right)\right]^2. \tag{7}$$

Here, w_{safety} is the corresponding scalar weight. This cost penalizes candidates whose minimum predicted separation falls below g_{safe}, and the penalty increases quadratically with the magnitude of the violation.

5) Overall Game-Aware Rollout Ranking: In addition to the game-aware cost, each sampled rollout is evaluated using a nominal quadratic tracking and control cost relative to a reference trajectory derived from the arc-length-parameterized raceline $\mathcal{G}$. For vehicle i, the reference progress $s_{i,t+\tau}^r$ is propagated recursively along $\mathcal{G}$ using the associated reference velocity profile $v^r(s)$, starting from $s_{i,t}^r = s_{i,t}$:

$$s_{i,t+\tau+1}^r = s_{i,t+\tau}^r + \Delta t \, v^r(s_{i,t+\tau}^r), \; \tau = 0, \ldots, H-1.$$

The corresponding reference states are obtained as $\mathbf{x}_{i,t+\tau}^r = \mathbf{x}^r(s_{i,t+\tau}^r)$. For each candidate rollout k, the nominal tracking and control cost is defined as:

$$\begin{aligned} c_{\text{track},i}^{(k)} = &\sum_{\tau=1}^{H} \left\| \mathbf{x}_{i,t+\tau}^{(k)} - \mathbf{x}_{i,t+\tau}^r \right\|_{\mathbf{Q}}^2 \\ &+ \sum_{\tau=0}^{H-1} \left\| \mathbf{u}_{i,t+\tau}^{(k)} \right\|_{\mathbf{R}}^2 + \sum_{\tau=1}^{H-1} \left\| \Delta \mathbf{u}_{i,t+\tau}^{(k)} \right\|_{\mathbf{S}}^2. \end{aligned} \tag{8}$$

Here $\Delta \mathbf{u}_{i,t+\tau}^{(k)} = \mathbf{u}_{i,t+\tau}^{(k)} - \mathbf{u}_{i,t+\tau-1}^{(k)}$, and $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{S}$ are weighting matrices. The total cost assigned to candidate k is:

$$c_i^{(k)} = c_{\text{track},i}^{(k)} + w_{\text{game}} c_{\text{game},i}^{(k)}. \tag{9}$$

Here, the weight w_{game} balances the influence of the game-aware cost against the nominal tracking and control cost.

3) 阻挡对齐成本：阻挡对齐项鼓励候选轨迹，使其横向位置与预测的对手 j 在预测范围末端的位置对齐。令 $\mathcal{T}_{\text{tail}}$ 表示预测范围最终 μ_{tail} 部分内的范围索引集。候选 k 和对手 j 在此预测范围最终部分上的横向偏移量的均值定义如下：

$$\bar{d}_i^{(k)} = \frac{1}{|\mathcal{T}_{\text{tail}}|} \sum_{\tau \in \mathcal{T}_{\text{tail}}} d_{i,t+\tau}^{(k)}, \; \bar{d}_j = \frac{1}{|\mathcal{T}_{\text{tail}}|} \sum_{\tau \in \mathcal{T}_{\text{tail}}} \hat{d}_{j,t+\tau},$$

ˆ其中 $d_{j,\ t+\tau}$ 表示对手 j 的预测横向偏移量。相应的尾部横向间隙为：$\Delta d_{ij}^{(k)}{}_{,\ \text{尾部}} = \left| \bar{d}_i^{(k)} - \bar{d}_j \right|$ · 使用此间隙，阻挡对齐成本被定义为：

$$c_{\text{block},ij}^{(k)} = -\gamma_{\text{block},ij} \left(1 + \Delta d_{ij,\text{tail}}^{(k)}\right)^{-1}. \tag{5}$$

这里，$\gamma_{\text{block},\ ij} = w_{\text{block}} \mathbb{I}[\Delta s_{ij,t} > s_{\text{role}}] \, \mathbb{I}[\Delta s_{ij,t} < s_{\text{contest}}]$ 在车辆 i 具有至少 s_{role} 的纵向优势且仍处于竞争范围 s_{contest} 内时激活阻挡成本，其中 $\Delta s_{ij,\ t} = s_{i,\ t} - \hat{}s_{j,\ t}$。标量 w_{block} 是权重。这种横向对齐减少了尾随车辆的可用空间，并促进了防御性阻挡。

4) 安全成本：安全项惩罚那些与对手的预测分离距离低于期望阈值 g_{safe} 的候选轨迹。对于候选 k 和对手 j，最小时间对齐距离为：

$$g_{\min,ij}^{(k)} = \min_{\tau=1,\ldots,H} \left\| \mathbf{p}_{i,t+\tau}^{(k)} - \hat{\mathbf{p}}_{j,t+\tau} \right\|_2, \tag{6}$$

其中$\mathbf{p}_i^{(k)}{}_{,t+\tau}$ 和 $\hat{}\mathbf{p}_{j,t+\tau}$ 分别表示候选 k 关联的位置和对手 j 在时间$t+\tau$的预测位置。相应的成对安全成本为：

$$c_{\text{safety},ij}^{(k)} = w_{\text{safety}} \left[\max\left(0, g_{\text{safe}} - g_{\min,ij}^{(k)}\right)\right]^2. \tag{7}$$

这里，w_{safety} is the 对应的标量我们得到。该成本惩罚预测最小分离值低于 g_{safe}的候选，且惩罚随违规幅度的平方值增加。

5) 整体博弈感知滚动排序：除了博弈感知成本外，每个采样的滚动路径还会使用相对于从弧长参数化赛道 $\mathcal{G}$派生的参考轨迹的标称二次跟踪和控制成本进行评估。对于车辆 i，参考进度$s_{i,t+\tau}^r$ 沿着 $\mathcal{G}$ 使用关联的参考速度曲线 $v^r(s)$递归传播，从$s_{i,t}^r = s_{i,t}$开始：

$$s_{i,t+\tau+1}^r = s_{i,t+\tau}^r + \Delta t \, v^r(s_{i,t+\tau}^r), \; \tau = 0, \ldots, H-1.$$

相应的参考状态获得为$\mathbf{x}_{i,t+\tau}^r = \mathbf{x}^r(s_{i,t+\tau}^r)$。对于每个候选滚动路径 k，标称跟踪和控制成本定义为：

$$\begin{aligned} c_{\text{track},i}^{(k)} = &\sum_{\tau=1}^{H} \left\| \mathbf{x}_{i,t+\tau}^{(k)} - \mathbf{x}_{i,t+\tau}^r \right\|_{\mathbf{Q}}^2 \\ &+ \sum_{\tau=0}^{H-1} \left\| \mathbf{u}_{i,t+\tau}^{(k)} \right\|_{\mathbf{R}}^2 + \sum_{\tau=1}^{H-1} \left\| \Delta \mathbf{u}_{i,t+\tau}^{(k)} \right\|_{\mathbf{S}}^2. \end{aligned} \tag{8}$$

这里 $\Delta \mathbf{u}_i^{(k)}{}_{,t+\tau} = \mathbf{u}_i^{(k)}{}_{,t+\tau} - \mathbf{u}_i^{(k)}{}_{,t+\tau-1}$，以及 $\mathbf{Q}$、$\mathbf{R}$ 和 $\mathbf{S}$ 是权重矩阵。分配给候选 k 的总成本是：

$$c_i^{(k)} = c_{\text{track},i}^{(k)} + w_{\text{game}} c_{\text{game},i}^{(k)}. \tag{9}$$

这里，权重 w_{game} 平衡了游戏感知成本与标称跟踪和控制成本的影响。

B. Feasibility Selection

After batch cost evaluation, feasibility selection applies explicit track-boundary and inter-vehicle collision checks to the ranked candidates and selects the lowest-cost candidate that passes both checks.

1) Track-Boundary Check: The track-boundary clearance of candidate k at prediction step τ is defined as follows: $\rho^{(k)}_{\mathrm{bd},i,t+\tau} = \min\left(w^R(s^{(k)}_{i,t+\tau}) + d^{(k)}_{i,t+\tau},\ w^L(s^{(k)}_{i,t+\tau}) - d^{(k)}_{i,t+\tau}\right)$. The minimum boundary clearance over the prediction horizon is $\rho^{(k)}_{\mathrm{bd},i} = \min_{\tau=1,\dots,H} \rho^{(k)}_{\mathrm{bd},i,t+\tau}$. A candidate passes the track-boundary check if its minimum clearance remains above the required threshold m_{bd}:

$$b^{(k)}_i = \mathbb{I}\big[\rho^{(k)}_{\mathrm{bd},i} \geq m_{\mathrm{bd}}\big]. \tag{10}$$

2) Collision-Avoidance Check: The inter-vehicle collision check uses the time-aligned minimum-distance operator defined in (6). For candidate rollout k, the minimum clearance over all opponents is defined as follows: $\rho^{(k)}_{\mathrm{obs},i} = \min_{j\in\mathcal{I}_{-i}} g^{(k)}_{\mathrm{min},ij}$, where $g^{(k)}_{\mathrm{min},ij}$ is given by (6). The candidate passes the collision-avoidance check if this minimum clearance remains above the collision threshold d_{col}:

$$o^{(k)}_i = \mathbb{I}\left[\rho^{(k)}_{\mathrm{obs},i} \geq d_{\mathrm{col}}\right]. \tag{11}$$

After ranking the candidate trajectories in $\mathcal{C}^{\ell}_{i,t}$ according to the cost in (9), each candidate is marked as feasible if it passes both the track-boundary and collision-avoidance checks. The lowest-cost candidate that passes both feasibility checks is selected as the planned trajectory at IBR iteration ℓ:

$$k^*_i = \arg\min_{k:h^{(k)}_i=1} c^{(k)}_i, h^{(k)}_i = b^{(k)}_i\, o^{(k)}_i. \tag{12}$$

Feasibility selection complements cost-based rollout ranking: the composite cost ranks the candidates, whereas the boundary and collision checks determine which candidates satisfy the prescribed feasibility criteria. The selected control sequence is used to warm-start the next planning step. If no feasible candidate exists, a least-violation fallback strategy is invoked. The boundary and collision violations for each candidate are defined as: $\nu^{(k)}_{\mathrm{bd},i} = \max\big(0, m_{\mathrm{bd}} - \rho^{(k)}_{\mathrm{bd},i}\big)$, $\nu^{(k)}_{\mathrm{obs},i} = \max\big(0, d_{\mathrm{col}} - \rho^{(k)}_{\mathrm{obs},i}\big)$, with the total violation given by $\nu^{(k)}_i = \nu^{(k)}_{\mathrm{bd},i} + \nu^{(k)}_{\mathrm{obs},i}$. The fallback strategy selects the candidate with the smallest combined violation: $k^* = \arg\min_k \nu^{(k)}_i$.

IV. Results and Discussion

This section evaluates SGTP in long-duration, closed-loop competitive multi-vehicle racing scenarios. We compare SGTP with a range of representative baselines, conduct ablation studies to assess the contributions of the proposed modules, and validate its performance across challenging tracks. The evaluation focuses on competitiveness and real-time efficiency.

A. Literature Benchmarks

The proposed method is compared with **eight literature benchmarks**. These include a sampling-based lattice planner [27], the learning-based End2Race method in [9], and the conditional flow-matching (CFM) planner proposed in [10]. End2Race [9] predicts control commands from past velocities and LiDAR inputs, whereas the CFM planner [10] generates

4

B. 可行性选择

批量成本评估后，可行性选择会对排名候选者应用显式的赛道边界和车辆间碰撞检查，并选择通过两项检查的最低成本候选者。

1) 赛道边界检查：候选 k 在预测步 τ 的赛道边界间隙定义为如下：$\rho^{(k)}_{\mathrm{bd}\ i,t+\tau} = \min\left(w^R(s^{(k)}_{i},_{t+\tau}) + d^{(k)}_{i},_{t+\tau},\ w^L(s^{(k)}_{i},_{t+\tau}) - d^{(k)}_{i},_{t+\tau}\right)$ 预测范围内的最小边界间隙是 $\rho^{(k)}_{\mathrm{bd}\ i} = \min_{\tau=1,\dots,H} \rho^{(k)}_{\mathrm{bd}\ i,t+\tau}$。如果候选的最小间隙保持在所需阈值 m_{bd} 以上，则通过赛道边界检查：

$$b^{(k)}_i = \mathbb{I}\big[\rho^{(k)}_{\mathrm{bd},i} \geq m_{\mathrm{bd}}\big]. \tag{10}$$

2) 避撞检查：车辆间避撞检查使用在（6）中定义的时间对齐最小距离算子。对于候选轨迹 k，所有对手的最小间隙定义为如下：$\rho^{(k)}_{\mathrm{obs}\ i} = \min_{j\in\mathcal{I}_{-i}} g^{(k)}_{\mathrm{min}\ ij}$，其中 $g^{(k)}_{\mathrm{min}\ ij}$ 由（6）给出。如果该最小间隙保持在避撞阈值 d_{col}：以上，则通过避撞检查：

$$o^{(k)}_i = \mathbb{I}\left[\rho^{(k)}_{\mathrm{obs},i} \geq d_{\mathrm{col}}\right]. \tag{11}$$

在根据$\mathcal{C}_{i,\ t}$ 根据（9）中的成本对候选轨迹进行排序后，如果每个候选轨迹都通过赛道边界和避撞检查，则将其标记为可行。通过这两个可行性检查的最低成本候选轨迹将在IBR迭代 ℓ：

$$k^*_i = \arg\min_{k:h^{(k)}_i=1} c^{(k)}_i, h^{(k)}_i = b^{(k)}_i\, o^{(k)}_i. \tag{12}$$

可行性选择补充了基于成本的滚动排序：复合成本对候选轨迹进行排序，而边界和避撞检查则确定哪些候选轨迹满足规定的可行性标准。选定的控制序列用于为下一步规划提供预热启动。如果没有可行的候选轨迹，则调用最小违规回退策略。每个候选轨迹的边界和避撞违规定义为：$\nu^{(k)}_{\mathrm{bd}\ i} = \max(0,\ m_{\mathrm{bd}} - \rho^{(k)}_{\mathrm{bd}\ i})$，$\nu^{(k)}_{\mathrm{obs}\ i} = \max(0\ d_{\mathrm{col}} - \rho^{(k)}_{\ i})$ $\nu^{(k)}_i =$ ， obs，总违规为$_{\mathrm{bd}\ i} + \nu^{(k)}_{\mathrm{obs}\ i}$。回退策略选择综合违规最小的候选轨迹：$k^* = \arg\min_k \nu^{(k)}_i$.

IV. 结果与讨论

本节评估 SGTP 在长时、闭环竞争性多车辆赛车场景中的表现。我们将 SGTP 与一系列代表性基线进行比较，进行消融研究以评估所提出模块的贡献，并在具有挑战性的赛道上验证其性能。评估重点关注竞争力和实时效率。

A. 文献基准

所提出的方法与**八项文献基准**进行了比较。这些包括基于采样的格点规划器 [27], 以及 [9], 中基于学习的 End2Race 方法，还有 [10] 中提出的条件流匹配（CFM）规划器。End2Race [9] 根据过去的速度和 LiDAR 输入预测控制指令，而 CFM 规划器 [10] 生成

plans conditioned on opponent predictions and the raceline. We also compare SGTP with the Race Stack framework in [2], a rule-based FSM method validated in real-world racing, and with the EVO-MPCC method proposed in [28], which has also been validated in real-world racing scenarios. In addition, we include two MPPI-based baselines: standard MPPI [22] and Biased-MPPI [23]. Finally, following [20], we integrate EVO-MPCC into the IBR loop to compare SGTP with an MPC-based solver under the same IBR framework.

Two ablation variants are evaluated. The first ablation, named SGTP w/o GC, removes the game-aware cost term in (2) while retaining feasibility selection. The second, named GA-IBR-MPPI, incorporates the same game-aware cost into the IBR loop but uses the standard MPPI formulation, which updates the nominal control sequence through path-integral importance weighting [22]. In all cases, the planned trajectories are tracked with the same pure-pursuit controller [9]. For simulation, we use the F1TENTH Gym environment described in [9] and the vehicle model presented in [25].

B. Implementation and Racing Scenarios

SGTP and the MPPI baselines use K=128 samples over a 1.2s prediction horizon (H=12), with $\sigma_a = 0.335$ and $\sigma_\delta = 0.025$. The tracking-cost weights are $\mathbf{Q}$=$[60, 60, 47.75, 39.48]$, $\mathbf{R}$=$[8.43, 20.0]$, and $\mathbf{S}$=$[1.0, 19.26]$. We set w_{game} =60, s_{contest}=8.0m, s_{role}=1.0m, μ_{tail}=0.3, and g_{safe}=1.0m. We use L_{IBR}=2 IBR iterations, m_{bd} is 0.515m, and d_{col} is 0.9m. The game-aware cost weights $[w_{\text{contest}}, w_{\text{long}}, w_{\text{block}}, w_{\text{safety}}]$ = $[1.0, 2.0, 10.0, 50.0]$ are manually tuned. The MPC-based benchmark employs the CasADi [29] software tool, with the IPOPT solver. We run our SGTP, the MPPI and learning-based baselines on an NVIDIA GeForce RTX 4060 GPU. We employ the following 7 racetracks from the open-source MapZoo[a] repository: *warehouse_v1*, *Berlin*, *f-shape*, *Brands Hatch*, *Oschersleben*, *MoscowRaceway*, and *Nuerburgring*. For each track, the minimum-curvature raceline [26] is used as the reference trajectory $\mathcal{G}$ for all vehicles. On each track, vehicles are initialized in a longitudinal queue along the track with a uniform 2m inter-vehicle spacing in ascending Frenet progress, with the ego vehicle always placed rearmost. The reference speeds of the opponent vehicles are scaled by a factor of 0.9 to promote close-range interactions.

C. Evaluation Metrics

A trial is considered a win if the ego vehicle achieves a greater Frenet progress gain (along the track reference line) than all opponents. Specifically, the win indicator is defined as $\mathbb{I}\left[\Delta s_e > \max_{j \neq e} \Delta s_j + \epsilon_s\right]$, where $\Delta s_j = s_{j,T} - s_{j,0}$ is the progress gain of vehicle j, Δs_e is the ego progress gain, T is the trial duration, and $\epsilon_s = 5 \times 10^{-3}$ is a small margin. The win rate over all trials is reported as *Wins*. We also monitor the Collision-Free Win rate (*CFW*) among winning trials; a higher value indicates cleaner wins without collisions. Since the ego vehicle is initialized behind the opponents in each trial, we further report the Final Pass Ratio (*FPR*) to quantify final overtaking performance. *FPR* is defined as the fraction of opponents overtaken by the ego vehicle by the end

[a]https://github.com/zhouhengli/MapZoo

基于对手预测和赛车路线的计划。我们还比较 SGTP 与中基于规则的 FSM 方法（已在真实赛车中验证），以及中提出的 EVO-MPCC 方法（该方法的验证场景也包含真实赛车）。此外，我们包含两个基于 MPPI 的基线：标准 MPPI [22]和 Biased-MPPI [23]。最后，遵循 [20]，我们将 EVO-MPCC 集成到 IBR 循环中，以在相同的 IBR 框架下比较 SGTP 与基于 MPC 的求解器。

评估了两种消融变体。第一种消融变体，命名为 SGTP 无 GC，移除了（2）中的博弈感知成本项，同时保留了可行性选择。第二种，命名为 GA-IBR-MPPI，将相同的博弈感知成本整合到 IBR 循环中，但使用标准的 MPPI 公式，该公式通过路径积分重要性加权更新名义控制序列 [22]。在所有情况下，规划的轨迹均使用相同的纯追踪控制器 [9]进行跟踪 [9]。对于仿真，我们使用 [9]中描述的 F1TENTHGym 环境，以及 [25]中提出的车辆模型。

B. 实施与赛车场景

SGTP 和 MPPI 基线在 1.2 秒预测范围（H=12）内使用 K=128 个样本，$\sigma_a = 0.335$ 和 $\sigma_\delta =$ 为 0.025。跟踪成本权重为 $\mathbf{Q}$=$[60, 60, 47.75, 39.48]$,$\mathbf{R}$=$[8.43, 20.0]$, 和 $\mathbf{S}$ =$[1.0, 19.26]$。我们设置 w_{game} =60,s_{contest}=8 为 0.0m, s_{role}=1 为 0.0m，μ_{tail}=0 为 0.3，以及 g_{safe}=1 为 0.0m。我们使用 L_{IBR}=2 个 IBR 迭代，m_{bd} 为 0.515m，d_{col} 为 0.9m。博弈感知成本权重 $[w_{\text{contest}}, w_{\text{long}}, w_{\text{block}}, w_{\text{safety}}]$ = $[1.0, 2.0, 10.0, 50.0]$ 由人工手动调整。基于 MPC 的基准采用 CasADi [29]软件工具，并使用 IPOPT 求解器。我们在 NVIDIA GeForce RTX 4060 GPU 上运行 SGTP、MPPI 和基于学习的基准。我们采用开源 MapZoo[a] 仓库中的以下 7 条赛道：*warehouse v1*,柏林, *f-shape*, *BrandsHatch*, *Oschersleben*,*MoscowRaceway*, 和 *Nuerburgring*。对于每条赛道，最小曲率赛车线 [26] 被用作所有车辆的参考轨迹 $\mathcal{G}$。在每条赛道上，车辆沿赛道以纵向队列初始化，车辆间距为 2m，在升序弗内特进度中保持均匀，自我车辆始终放置在最后方。对手车辆的参考速度按 0.9 的因子缩放，以促进近距离交互。

C. 评估指标

如果自我车辆在赛道参考线上的弗内特进度增益（Frenet progress gain）大于所有对手，则该试验被视为获胜。具体来说，胜指标定义为 $\mathbb{I}\left[\Delta s_e > \max_{j \neq e} \Delta s_j + \epsilon_s\right]$, 其中 $\Delta s_j = s_{j,T} - s_{j,0}$ 是车辆 j, Δs_e 是自我进度增益，T是试验持续时间，而 $\epsilon_s = 5 \times 10^{-3}$ 是一个小的余量。所有试验的胜率以 *Wins* 报告。我们还监测获胜试验中的无碰撞胜率（*CFW*）；值越高表示无碰撞的更干净的胜利。由于自我车辆在每个试验中初始化在对手后面，我们进一步报告最终超越比率（*FPR*）以量化最终超越性能。*FPR* 定义为试验结束时自我车辆超越的对手的比例：FPR $\mathbb{I}[\Delta s_e >$，其中 $_{j \neq e} \Delta s_j + \epsilon_s$]是车辆数量。

[a]https://github.com/zhouhengli/MapZoo

of the trial: FPR $= \frac{1}{n_v-1}\sum_{j\neq e}\mathbb{I}\left[s_{e,T} > s_{j,T} + \epsilon_s\right]$, where n_v is the number of vehicles. To assess long-duration racing performance, we report the maximum trial duration $D_{\max}$ and the mean trial duration D_{mean}. A trial terminates when either a collision occurs or the predefined maximum duration is reached. The Close-interaction Segment Duration (*CSD*) measures the mean duration of segments during which an opponent remains within 1.0m longitudinally and 0.5m laterally of the ego vehicle in Frenet coordinates. The mean ego velocity, denoted as $\bar{v}_{\text{ego}}$, is computed by averaging the ego speed over all samples from all trials. Mean Control Smoothness is reported by *MCS* and is computed as $\frac{1}{T}\sum_{k=1}^{T}\|\Delta\mathbf{u}_k\|_2^2$. Computation Time (*CT*), defined as the elapsed wall-clock time of each planning call, is used to evaluate real-time efficiency. For IBR-based game planners, *CT* corresponds to the total time over all IBR iterations and the additional ego response. We report CT_{mean}, CT_{std}, and $CT_{\max}$ over all valid trials for statistical comparison. In the radar visualization of Fig. 5, the six normalized axes represent long-duration interaction (D_{mean}), winning ability (*Wins*), overtaking ability (*FPR*), close-range interaction (*CSD*), aggressiveness ($\bar{v}_{\text{ego}}$), and runtime efficiency (the inverse-normalized CT_{mean}).

D. Qualitative Evaluation of Multi-Behavior Interactions

Three challenging tracks, *Berlin*, *f-shape*, and *Brands Hatch*, are used to evaluate SGTP in highly interactive racing scenarios. Each trial lasts up to 40s and includes three vehicles: the ego vehicle, Opponent 1 (O1), and Opponent 2 (O2). All vehicles are controlled by SGTP, and the opponents respond reactively and competitively to the ego vehicle during the race. As shown in Fig. 3 and 4, SGTP produces sustained close-range interactions in multi-vehicle racing. On the *Berlin* track, O1 (blue) executes a defensive blocking maneuver during time t_1–t_2 by deviating from the raceline (zoomed view (a)). This maneuver is to defend its position and prevent the ego vehicle (red) from overtaking. The ego vehicle responds by trailing O1 at t_2–t_4 and then uses its speed advantage (Fig. 4) to engage sustained wheel-to-wheel contesting at t_4–t_7, progressively securing a favorable racing corridor and ultimately achieving an opportunistic overtake at t_8, as also shown in Fig. 4. On the *f-shape* track (Fig. 3, middle), O1 initiates a defensive blocking maneuver at t_2 (see (b) zoom-in view). As the interaction evolves, O1 moves farther from the raceline. The trade-off between blocking and raceline tracking then favors its return toward the raceline between t_3 and t_5. The ego preserves an overtaking corridor, maintains feasible wheel-to-wheel racing, and completes the overtake from t_5 to t_8. On the *Brands Hatch* track, the ego vehicle simultaneously faces blocking from O2 ahead and overtaking pressure from O1 behind at t_2 (see (c) zoom-in view). Nevertheless, it completes an opportunistic overtake during t_3–t_4 by exploiting its speed advantage.

These results show that SGTP generates diverse competitive behaviors through IBR updates and game-aware rollout ranking, including blocking, wheel-to-wheel contesting, and opportunistic overtaking. The results further show that SGTP can transition smoothly among behaviors as interactions evolve.

为评估长时赛车性能，我们报告最大试验持续时间 $D_{\max}$ 和平均试验持续时间 D_{mean}。当发生碰撞或达到预定义的最大持续时间时，试验终止。接近交互段持续时间（CSD）测量对手在弗内特坐标中纵向保持1.0m、横向保持0.5m内自我车辆的平均持续时间。平均自我速度 $\bar{v}_{\text{ego}}$ 是通过对所有试验的所有样本的自我速度进行平均计算的。平均控制平滑度由 MCS 报告，并计算为 $\frac{1}{T}\sum_{k=1}^{T}\|\Delta u_k\|_2^2$.计算时间（CT）定义为每个规划调用的经过的墙上时间，用于评估实时效率。对于基于 IBR 的游戏规划器，CT 对应于所有 IBR 迭代和额外的自我响应的总时间。我们报告 CT_{mean}，CT_{std} 和 $CT_{\max}$ 以所有有效试验进行统计比较。在图 5 的雷达可视化中，六个标准化轴表示长时交互 (D_{mean})、获胜能力（Wins）、超越能力（FPR）、近距离交互（CSD）、侵略性 ($\bar{v}_{\text{ego}}$) 和运行时效率（逆标准化 CT_{mean}）。

D. 多行为交互的定性评估

使用柏林、f形和 Brands Hatch 三个具有挑战性的赛道，评估 SGTP 在高度交互的赛车场景中的表现。每次试验持续最长 40 秒，包括三辆车：自我车辆、对手1（O1）和对手2（O2）。所有车辆均由 SGTP 控制，对手在比赛中对自我车辆做出反应性和竞争性的回应。如图 3 和 4 所示，SGTP 在多车辆赛车中产生持续的近距离交互。在柏林赛道上，O1（蓝色）在 t_1–t_2 期间执行防御性阻挡动作，通过偏离赛车线（放大视图（a））， 目的是保卫其位置并阻止自我车辆（红色）超越。自我车辆通过在 t_2–t_4 期间跟随 O1 做出回应，然后利用其速度优势（图 4）在 t_4–t_7 期间参与持续的 车对车 争斗，逐步占据有利的赛车走廊，并最终在 t_8 实现机会主义超越，如图 4 所示。在 f形赛道上（图 3，中间），O1 在 t_2 （参见（b）放大视图）发起防御性阻挡动作。随着交互的演变，O1 越来越远离赛车线。阻挡和赛车线跟踪之间的权衡使其在 t_3 和 t_5 期间回归赛车线。自我车辆保持超越走廊，维持可行的 车对车 赛车，并在 $t_5 - t_8$ 完成超越。在 Brands Hatch 赛道上，自我车辆在 t_2 （参见（c）放大视图）同时面临来自前方 O2 的阻挡和后方 O1 的超越压力。尽管如此，它通过利用其速度优势，在 t_3–t_4 期间完成了机会主义超越。

这些结果表明，SGTP通过IBR更新和游戏感知的 rollout 排名生成多样化的竞争行为，包括阻挡、车轮对车轮的竞争以及机会主义的超越。结果进一步表明，随着交互的演变，SGTP可以在行为之间平稳过渡。

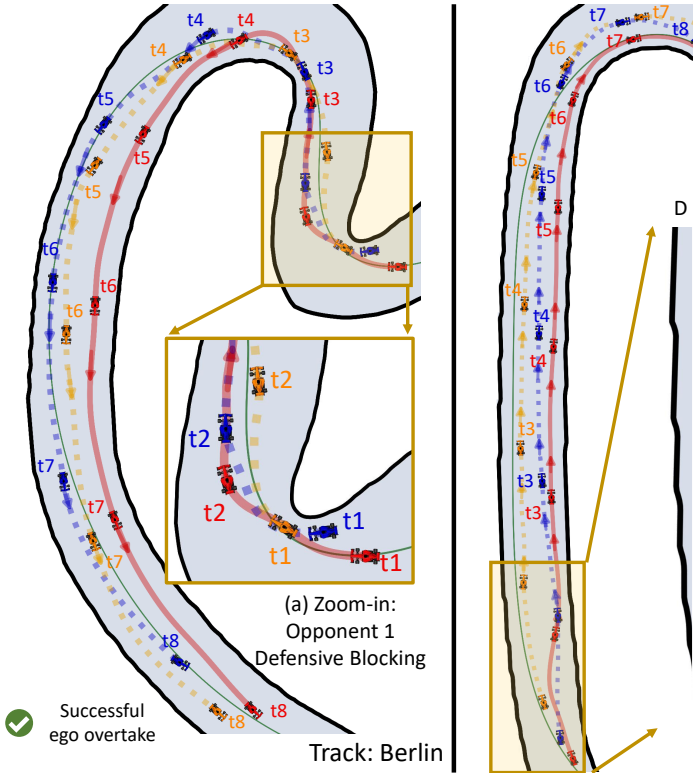


Fig. 3: **SGTP enables competitive behaviors and reliable tra multiple racetracks.** The ego vehicle demonstrates diverse racin before completing an opportunistic overtake after sustained close-r

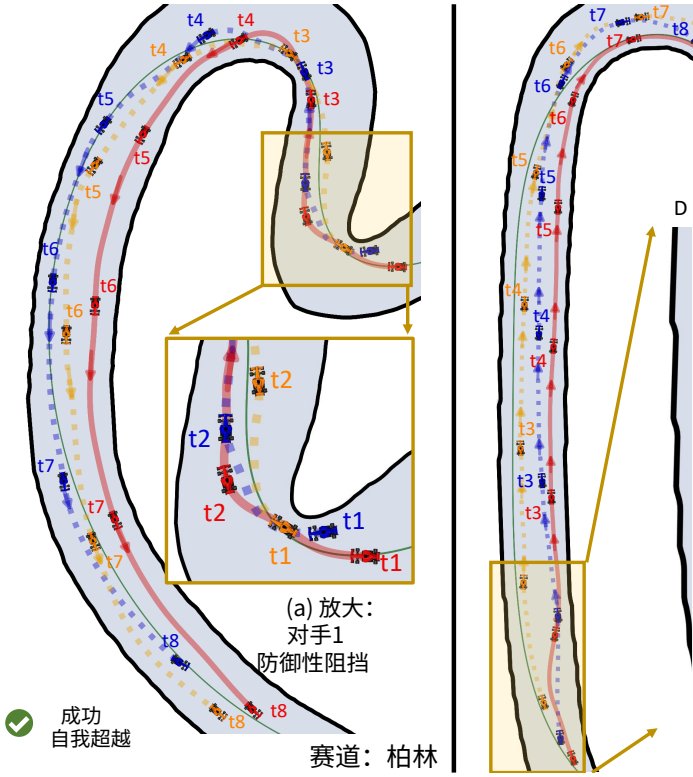


图3：**SGTP能够实现竞争性行为并在多条赛道上可靠地行驶。** 自我车辆在完成一次机会主义超车前展示了多样化的赛车行为，并在持续近距离追逐后…

E. Quantitative Results: Performance & Real-Time Efficiency

For each of the 7 tracks in Section IV-B, we sample 6 starting positions, yielding 42 trials with one ego and two opponents. Each trial runs for at most 50s and terminates if the ego vehicle is involved in a collision. The CFM and End2Race benchmarks use MPPI-controlled opponents for stable opponent predictions. For all other methods, the ego vehicle and opponents use the same planner.

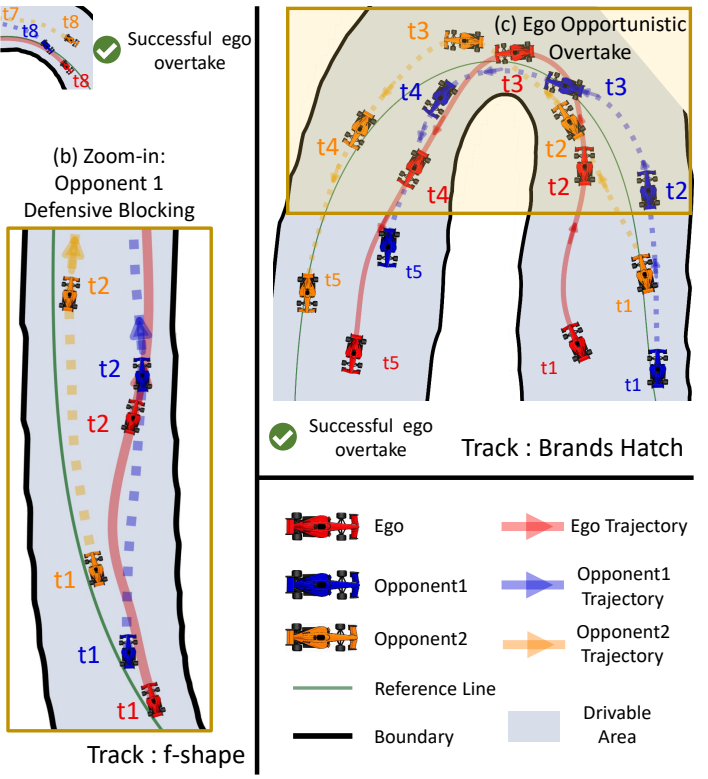
Our SGTP achieves the best overall performance (Table I), with **95.24%** *Wins* and **100.00%** wins without collisions (*CFW*). It also achieves $D_{\text{mean}} = 49.67$s under a predefined maximum trial duration of 50s, indicating that it can sustain racing for nearly the entire allowed duration. In addition, our SGTP obtains the highest *FPR* of **75.00%** and *CSD* of **0.99**, indicating successful overtaking after sustained close-range interactions. Meanwhile, it maintains competitive speed ($\bar{v}_{\text{ego}} = 5.86$ m/s) and smooth controls (*MCS* = 0.008). As shown in Fig. 5, SGTP presents the most balanced profile across long-duration racing, winning ability, overtaking ability, tense interaction, aggressiveness, and runtime efficiency. It also runs in real time, with $CT_{\text{mean}} = \mathbf{0.095}$ s (Table I). Compared with the other benchmarks, only EVO-MPCC and IBR-MPC achieve relatively high win rates of 92.86% and 85.71%, respectively. However, neither method can sustain intense competitive interactions over extended racing periods. In addition, their planning processes are both time-consuming and unstable. Specifically, the CT_{mean} values for EVO-MPCC and IBR-MPC are 0.860s and 1.903s, while the corresponding CT_{std} values are 0.646s and 1.043s, making these methods

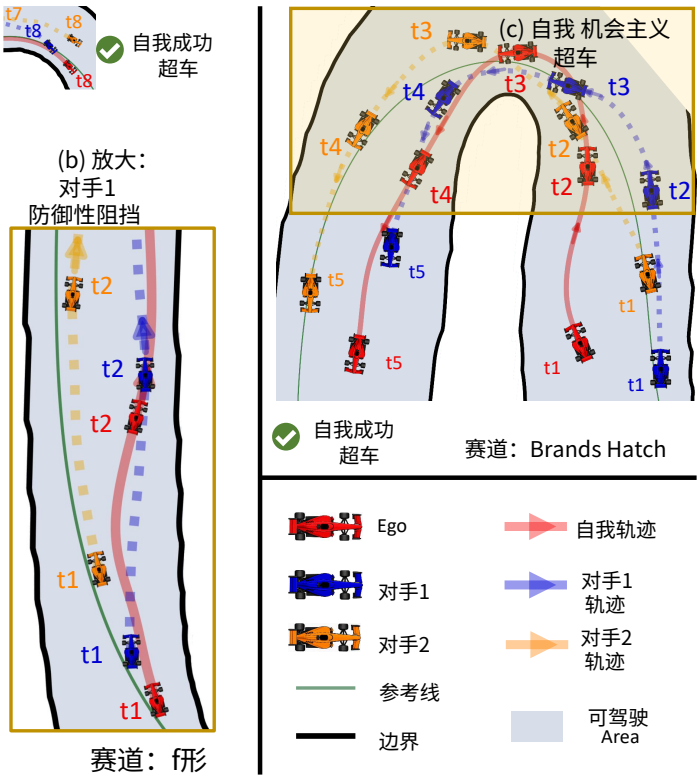
E. 定量结果：性能与实时效率

对于第IV-B节中的7条赛道，我们采样6个起始位置，产生42次试验，每次试验包含一辆自我车辆和两辆对手。每次试验最多运行50秒，如果自我车辆发生碰撞则终止。CFM和End2Race基准使用MPPI控制的对手以实现稳定的对手预测。对于所有其他方法，自我车辆和对手使用相同的规划器。

我们的SGTP实现了最佳的整体性能（表I），具有**95.24%**胜率和**100.00%** 无碰撞胜率（基于累积*Fren b*的碰撞超越）。它还在预定义的最大试验时长为50秒的情况下实现了 $D_{\text{mean}} = 49.67$s，表明它可以在几乎整个允许的时间内持续赛车。此外，我们的SGTP获得了最高的 *FPR* 为**75.00%** 和*CSD* 为**0.99**，表明在持续的近距离交互后成功超越。同时，它保持了竞争性速度（$\bar{v}_{\text{ego}} = 5.86$ m/s）和平滑控制（*MCS* = 0.008）。如图5所示，SGTP在长时赛车、获胜能力、超越能力、紧张交互、侵略性和运行时效率方面呈现出最平衡的轮廓。它还实时运行，具有 $CT_{\text{mean}} = \mathbf{0.095}$s（表I）。与其他基准相比，只有EVO-MPCC和IBR-MPC实现了相对较高的胜率，分别为92.86%和85.71%。然而，这两种方法都不能在长时间的赛车过程中维持激烈的竞争交互。此外，它们的规划过程既耗时又不稳定。具体来说，EVO-MPCC和IBR-MPC的 CT_{mean} 值为0.860秒和1.903秒，而相应的 CT_{std} 值为0.646秒和1.043秒，这使得这些方法

6





transitions between them in highly interactive scenarios on cing behaviors, including trailing and wheel-to-wheel contesting, se-range interactions. Videos are available on the project webpage.

在高度交互场景中，它们之间的过渡包括跟驰和车距竞争等交互行为。视频在项目网页上找到。

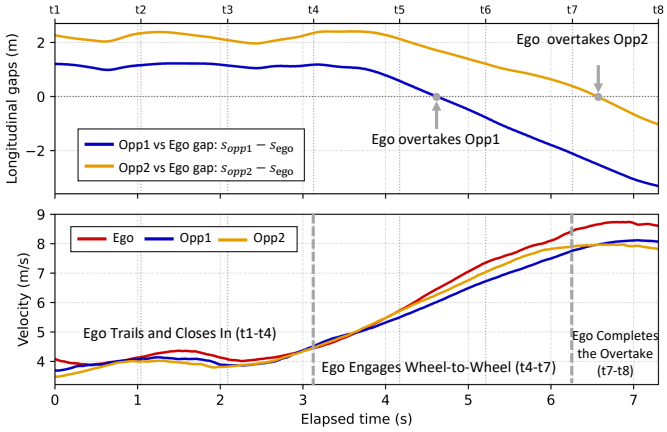


Fig. 4: Longitudinal gaps and velocity profiles from t_1 to t_8 on the *Berlin* track (Fig. 3, left), where the SGTP-controlled ego vehicle overtakes two SGTP-controlled opponents.

unsuitable for real-time planning in rapidly changing multi-vehicle racing scenarios. The ablation results verify the importance of our two key modules: game-aware cost and feasibility selection. Without the game-aware cost, SGTP w/o GC drops to 50.00% *Wins* and 14.29% *CFW*, indicating weaker competitive performance. Without the feasibility selection, GA-IBR-MPPI achieves only 57.14% *Wins* and 0.00% *CFW*, while its $D_{\max}$ is only 5.21s, showing that feasibility selection is critical for long-duration competitive racing.

F. Defensive Capability and Robustness to Perturbations

Across the 7 tracks in Section IV-B, we sample 15 positions per track, yielding 105 trials to evaluate SGTP's defensive

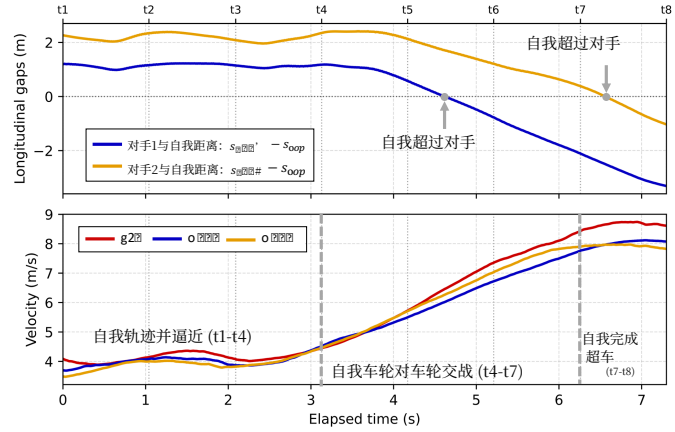


图4：从 t_1 到 t_8在柏林赛道上（图3，左侧），SGTP控制的自我车辆超过两个SGTP控制的对手。

不适用于快速变化的多车辆赛车场景中的实时规划。消融实验结果验证了我们两个关键模块的重要性：博弈感知成本和可行性选择。没有博弈感知成本，SGTP 无 GC 降至 50.00 % 胜率和 14.29% 基于累积*Fren b*的碰撞超越，表明竞争力较弱。没有可行性选择，GA-IBR-MPPI 仅达到 57.14% 胜率和 0.00% 基于累积*Fren b*的碰撞超越，而其$D_{\max}$ 仅为 5.21s，表明可行性选择对于长时竞争性赛车至关重要。

F. 防御能力和对扰动的鲁棒性

跨越7条赛道在第四部分B中，我们每条赛道采样15个位置，产生105次试验来评估SGTP的防御

TABLE I: Quantitative Comparison and Ablation Study of Multi the Proposed SGTP in Long-duration Competitive Racing and R

Methods		Wins[a] ↑	CFW↑	Long-dur FPR[a]	Long-dur CSD[b]
Sampling	Lattice [27]	50.00%	0.00%	20.24%	0.23
FSM	Race Stack [2]	45.24%	5.26%	8.33%	0.20
Learning	End2Race [9]	66.67%	0.00%	11.90%	0.05
	CFM [10]	38.10%	0.00%	0.00%	0.02
Predictive Control	MPPI [22]	4.76%	0.00%	3.57%	0.00
	Biased-MPPI [23]	2.38%	0.00%	2.38%	0.00
	EVO-MPCC [28]	92.86%	2.56%	8.33%	0.08
Game	IBR-MPC [20]	85.71%	5.56%	10.71%	0.07
	SGTP w/o GC	50.00%	14.29%	28.57%	0.14
	GA-IBR-MPPI	57.14%	0.00%	19.05%	0.02
	SGTP (ours)	**95.24%**	**100.00%**	**75.00%**	**0.99**

Bold and underlined values indicate the best and second-best results, respect
[a] *Wins* denotes the progress-based win rate, and *CFW* denotes the collision-fr overtaken by the ego vehicle by the end of a trial based on cumulative Fren
[b] *CSD* measures the mean duration of close-interaction segments. In conju interactions are sustained over longer periods.
[c] D_{mean} and $D_{\max}$ quantify the durations of sustained racing interaction $D_{\text{mean}}/$(maximum trial time) to represent the race completion ratio. MCS d

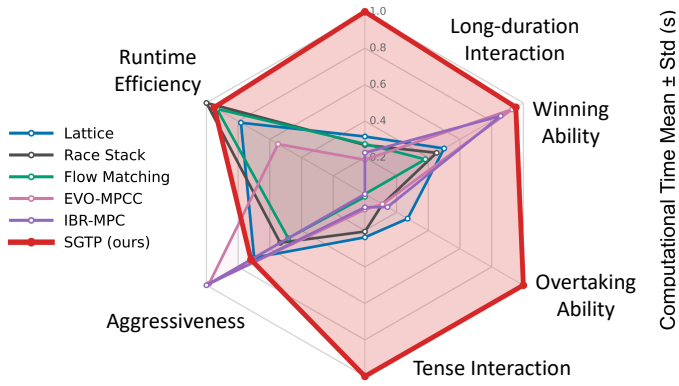


Fig. 5: **Competitive performance & scalability of SGTP.** Left p metrics of Sec. IV-C. Right plot: SGTP maintains low, consistent of vehicles increases, outperforming IBR-MPC [20]. Our webpa

capability and robustness to prediction perturbations (Table II). To evaluate defensive capability, the rearmost vehicle uses EVO-MPCC [28], while both opponents use SGTP, and their performance is reduced by 10% to complicate the defense. Only the EVO-MPCC agent performs the additional final response. Despite this advantage, its CFW is only 3.88%, demonstrating that SGTP can effectively defend against an MPC-based planner without performing the final response.

We next evaluate SGTP using perturbed opponent-trajectory predictions generated by the perturbation model in [30]. In this evaluation, the two opponents and the ego vehicle (starting from behind) are controlled by SGTP. Under mild perturbations, D_{mean} decreases by only 0.52%, while *CSD* decreases by 55.88%, indicating more conservative interactions. This results from the safety cost and feasibility selection, which penalize or reject candidates with insufficient clearance. Under severe perturbations, *CSD* and D_{mean} decrease by 63.86% and 4.41%, respectively. Nevertheless, SGTP maintains a CFW of 94.12%, demonstrating its capability to avoid collisions even with noisy opponents' predictions.

TABLEI：定量比较与消融研究——多车场景下长时竞速中提出的SGTP

方法		基于进度的胜率[a] ↑	CFW↑	长时 FPR[a]	长时 CSD[b]
采样	格网 [27]	50.00%	0.00%	20.24%	0.23
FSM	种族堆叠 [2]	45.24%	5.26%	8.33%	0.20
学习	End2Race [9]	66.67%	0.00%	11.90%	0.05
	CFM [10]	38.10%	0.00%	0.00%	0.02
预测控制	MPPI [22]	4.76%	0.00%	3.57%	0.00
	Biased-MPPI [23]	2.38%	0.00%	2.38%	0.00
	EVO-MPCC [28]	92.86%	2.56%	8.33%	0.08
Game	IBR-MPC [20]	85.71%	5.56%	10.71%	0.07
	SGTP 无 GC	50.00%	14.29%	28.57%	0.14
	GA-IBR-MPPI	57.14%	0.00%	19.05%	0.02
	SGTP（我们）	**95.24%**	**100.00%**	**75.00%**	**0.99**

加粗和下划线值表示最佳和次优结果，respect[a] *Wins* 表示基于进度的胜率，而*CFW* 表示试验结束时自我车辆通过累积Fren [b] *CSD* 测量接近交互段的平均持续时间。在conju交互会持续更长时间。

[c] D_{mean} 和 $D_{\max}$ 量化了持续赛车交互的持续时间$D_{\text{mean}}/$(最大试验时间)，以表示比赛完成比率。MCSd

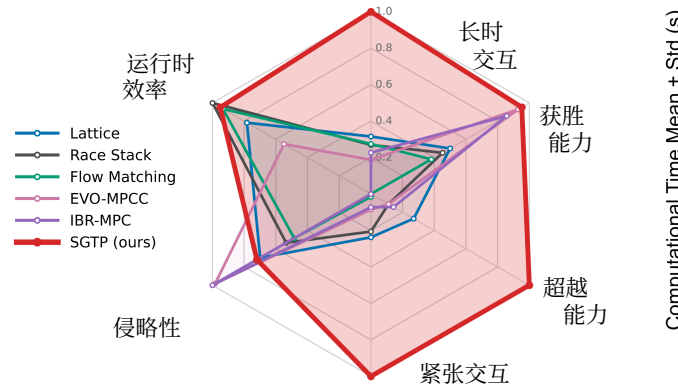


图5：**SGTP的竞争性能与可扩展性**。左图：第IV-C节中的p指标。右图：随着车辆数量增加，SGTP 保持低且一致的性能，优于 IBR-MPC [20]。我们的网页

能力和鲁棒性用于预测扰动（表 II）。为了评估防御能力，最末尾的车辆使用EVO-MPCC [28]，，而两个对手使用SGTP，它们的性能降低了10%以使防御复杂化。只有EVO-MPCC代理执行了额外的最终响应。尽管有这个优势，它的CFW只有3.88%，表明SGTP可以有效地防御基于MPC的规划器，而无需执行最终响应。

接下来，我们使用扰动模型在 [30] 中生成的扰动对手轨迹预测来评估 SGTP。在此评估中，两个对手和自我车辆（从后面开始）由 SGTP 控制。在轻微扰动下，D_{mean} 仅降低了 0.52%，而 *CSD* 降低了 55.88%，表明交互更加保守。这是由于安全成本和可行性选择，它们会惩罚或拒绝空间不足的候选方案。在严重扰动下，*CSD* 和 D_{mean} 分别降低了 63.86% 和 4.41%。尽管如此，SGTP 仍保持了 94.12% 的 CFW，展示了其即使面对噪声的对手预测也能避免碰撞的能力。

7

ulti-Vehicle Racing Planners, Demonstrating the Effectiveness of d Real-Time Planning.

多车辆竞速规划器，展示实时规划的有效性。

-duration Competitive Performance ↑						Real-Time Efficiency ↓		
D^b	D^c_{mean}	D^c_{ratio}	$D^c_{\max}$	$\bar{v}_{\text{ego}}$	$\text{MCS}^c \downarrow$	CT_{mean}	CT_{std}	$CT_{\max}$
3	16.73	33.46%	44.02	5.75	0.928	0.415	0.072	0.560
0	14.51	29.02%	**50**	4.90	0.037	-	-	-
5	2.84	5.69%	4.48	4.02	**0.002**	**0.003**	**3.9e-4**	**0.004**
2	14.78	29.55%	**50**	4.63	0.120	0.127	0.012	0.141
0	1.60	3.20%	5.15	2.16	0.004	0.041	0.001	0.042
0	1.36	2.73%	2.7	1.90	**0.002**	0.095	0.005	0.099
8	10.58	21.17%	**50**	7.23	0.011	0.860	0.646	3.743
7	12.46	24.91%	**50**	**7.32**	0.012	1.903	1.043	4.153
4	20.10	40.20%	**50**	4.87	0.006	0.116	0.006	0.123
2	2.22	4.44%	5.21	2.86	0.007	0.078	0.003	0.084
9	**49.67**	**99.35%**	**50**	5.86	0.008	0.095	0.004	0.102

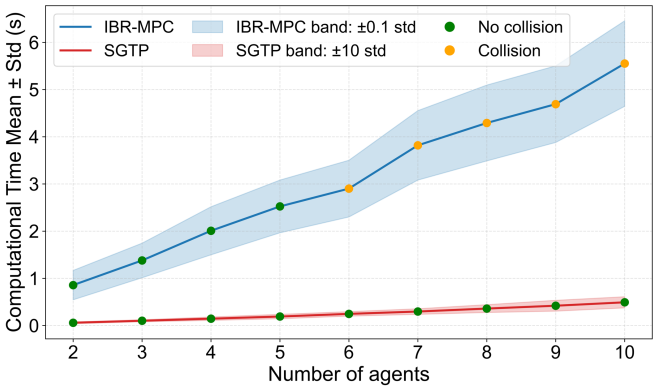
pectively; shading highlights SGTP. ↑/↓ denote higher/lower is better.
on-free win rate. *FPR* denotes the Final Pass Ratio, i.e., the fraction of opponents renet progress. Higher values are better for all three metrics.
onjunction with D_{mean}, a larger *CSD* indicates that high-intensity close-range

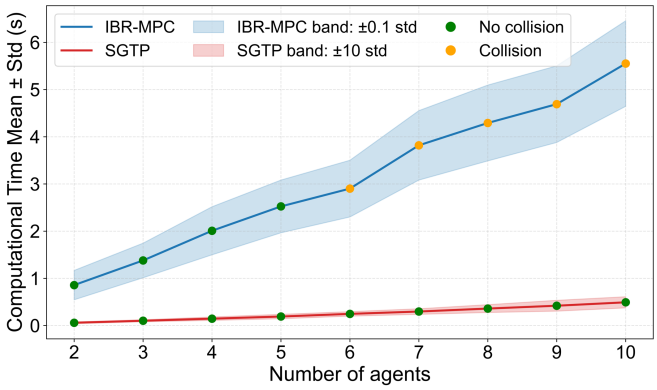
-持续时间 竞争性能 ↑						实时效率 ↓		
D^b	D^c_{mean}	$D^c_{\text{比率}}$	$D^c_{\max}$	$\bar{v}_{\text{ego}}$	$\text{MCS}^c \downarrow$	CT_{mean}	CT_{std}	$CT_{\max}$
3	16.73	33.46%	44.02	5.75	0.928	0.415	0.072	0.560
0	14.51	29.02%	**50**	4.90	0.037	-	-	-
5	2.84	5.69%	4.48	4.02	**0.002**	**0.003**	3.9e-4	**0.004**
2	14.78	29.55%	**50**	4.63	0.120	0.127	0.012	0.141
0	1.60	3.20%	5.15	2.16	0.004	0.041	0.001	0.042
0	1.36	2.73%	2.7	1.90	**0.002**	0.095	0.005	0.099
8	10.58	21.17%	**50**	7.23	0.011	0.860	0.646	3.743
7	12.46	24.91%	**50**	**7.32**	0.012	1.903	1.043	4.153
4	20.10	40.20%	**50**	4.87	0.006	0.116	0.006	0.123
2	2.22	4.44%	5.21	2.86	0.007	0.078	0.003	0.084
9	**49.67**	99.35%	**50**	5.86	0.008	0.095	0.004	0.102

分别；阴影突出显示SGTP。 ↑/↓表示更好/更差。在无干扰获胜率方面。
FPR 表示最终传球比率，即对手重新获得进球权（net progress）的比例。这三个指标中，数值越高越好。与 D_{mean}结合时，更大的*CSD*表明高强度近距离

tions, with larger values indicating longer interactions. D_{ratio} is computed as S denotes the mean control smoothness.

交互，且数值越大表示交互时间越长。 D_{ratio} 的计算方式为S表示平均控制平滑度。





ft plot: our SGTP achieves the best balance across the normalized ent computation times and collision-free operation as the number bpage shows videos with up to 10 agents.

ft图：我们的SGTP在归一化熵计算时间和无碰撞操作之间实现了最佳平衡，随着bpage数量的增加，展示了最多包含10个智能体的视频。

TABLE II: SGTP's capability to defend & block (first row) and robustness to opponents' prediction uncertainty (last 3 rows).

表II：SGTP防御与拦截能力（第一行）以及对手预测不确定性下的鲁棒性（最后三行）。

Methods	D_{mean}	ΔD_{mean}	CSD	ΔCSD	CFW
EVO-MPCC vs SGTP	12.00	-	0.06	-	3.88%
SGTP w/o noise	49.08	-	1.01	-	96.12%
SGTP mild noise	48.82	-0.52%	0.44	-55.88%	97.14%
SGTP severe noise	46.91	-4.41%	0.36	-63.86%	94.12%

方法	D_{mean}	ΔD_{mean}	CSD	ΔCSD	CFW
EVO-MPCC vs SGTP	12.00	-	0.06	-	3.88%
SGTP 无噪声	49.08	-	1.01	-	96.12%
SGTP 轻微噪声	48.82	-0.52%	0.44	-55.88%	97.14%
SGTP 严重噪声	46.91	-4.41%	0.36	-63.86%	94.12%

$\Delta D_{\text{mean}}, (\%)$ and $\Delta\text{CSD}, (\%)$ are computed relative to the corresponding values for SGTP w/o noise. The mild and severe settings use $(\sigma_{\text{lon}}, \sigma_{\text{lat}}) = (0.03, 0.015)$ and $(0.08, 0.04)$, respectively.

ΔD_{mean}，(%)和 ΔCSD，(%)是相对于SGTP无噪声时的对应值计算的。轻度设置和严重设置分别使用$(\sigma_{\text{lon}}$，$\sigma_{\text{lat}}) = (0.03$，$0.015)$和$(0.08$，$0.04)$。

G. Scalability Analysis Across Vehicle Counts

We now assess the scalability of SGTP as the number of interacting vehicles increases. On the *Brands Hatch* track, we vary the number of vehicles from 2 to 10, and each trial lasts 11s. As shown in Fig. 5, SGTP maintains low and stable computational time with small variance, while remaining collision-free across all vehicle counts. In contrast, IBR-MPC exhibits rapidly increasing computational time and greater variance, with collisions occurring in denser scenarios. Overall, SGTP combines IBR reasoning with sampling-based

G. 跨车辆数量可扩展性分析 *ts*

我们现在评估SGTP的可扩展性，随着交互车辆数量的增加。在*BrandsHatch* 赛道上，我们将车辆数量从2变化到10，每次试验持续11秒。如图5所示，SGTP保持低且稳定的计算时间，方差较小，同时在所有车辆数量下均保持无碰撞。相比之下，IBR-MPC的计算时间迅速增加且方差更大，在更密集的场景中会出现碰撞。总体而言，SGTP结合了IBR推理与基于采样的

planning to maintain real-time performance and generate reliable behaviors across varying numbers of vehicles.

V. Conclusions

This letter presented SGTP, a real-time game-theoretic planner for competitive multi-vehicle racing. By exploiting GPU-parallel sampling and rollout evaluation, SGTP efficiently approximates unilateral best responses in IBR. Feasibility selection explicitly enforces the track-boundary and collision-avoidance constraints, supporting reliable transitions between diverse behaviors. The key finding is that the game-aware cost can produce diverse competitive behaviors, including blocking, wheel-to-wheel contesting, and overtaking. Future work will reduce parameter tuning and incorporate world-model-based opponent style recognition to improve prediction.

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规划以保持实时性能，并在不同数量的车辆中生成可靠行为。

V. 结论

本信件介绍了SGTP，一种用于竞争性多车辆赛车的实时博弈规划器。通过利用GPU并行采样和 rollout 评估，SGTP高效地近似IBR中的单边最佳响应。可行性选择明确地执行赛道边界和避撞约束，支持在不同行为之间进行可靠转换。关键发现是，博弈感知成本可以产生多样化的竞争行为，包括阻挡、车轮对车轮的竞争和超越。未来的工作将减少参数调整，并结合基于世界模型的对手风格识别来提高预测能力。

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